

Rewired by time: a dopaminergic account of cognitive flexibility across the mouse lifespan

A thesis presented

by

Xiang Yi Sean Meng

Presented to

The Faculty of the Committee on Degrees in Neuroscience

in partial fulfillment of the requirements for the degree of

Bachelor of Arts

Harvard College

Cambridge, Massachusetts

March 13, 2026

The Harvard College Honor Code

Members of the Harvard College community commit themselves to producing academic work of integrity – that is, work that adheres to the scholarly and intellectual standards of accurate attribution of sources, appropriate collection and use of data, and transparent acknowledgement of the contribution of others to their ideas, discoveries, interpretations, and conclusions. Cheating on exams or problem sets, plagiarizing or misrepresenting the ideas or language of someone else as one's own, falsifying data, or any other instance of academic dishonesty violates the standards of our community, as well as the standards of the wider world of learning and affairs.

The Neuroscience Concentration AI Policy

Completing a thesis develops valuable skills in analytical thinking and scientific writing. **Students are strictly forbidden from including AI-generated content in any part of their thesis manuscript, including preliminary drafts.** Incorporating AI-produced text into your thesis constitutes an Honor Code violation at Harvard College and will be considered academic misconduct.

Certain applications of generative AI tools are acceptable during other research phases. For example, these tools may be valuable for broadening or deepening your knowledge of specific concepts, understanding methodologies, or troubleshooting analyses. However, exercise caution when using these resources: current AI models regularly produce inaccurate information. This is especially true for scientific research, as journals are often paywalled and thus under-represented in training data.

You bear full responsibility for the accuracy of all assertions in your thesis, making it essential to rigorously verify any AI contributions. You should be prepared to answer questions about any part of your thesis during the oral conference.

If you are unsure as to whether a particular use is allowed, contact your concentration advisor promptly.

By signing below, I confirm that I have adhered fully to all components of the Honor Code and AI Policy in the production of this thesis.

Digital Signature: _____

A handwritten signature in black ink, appearing to be 'G. M.', is written over a horizontal line.

Acknowledgments

I want to start by thanking my PI, Dr. Bernardo Sabatini. Bernardo, I remember first approaching you after a guest lecture to ask about the Kempner. I had no idea that conversation would lead to four years in your lab, and five to six more pursuing a PhD. Thank you for welcoming my wide-eyed freshman self into your lab and letting me pursue science to my heart's content. The freedom and trust you gave me so early changed the course of not only my time at Harvard but my whole career and life ahead. You are a wonderful mentor and a brilliant scientist. I have learned so much from you.

To Dr. Kevin Mastro, the mentor who I owe it all to, thank you for taking a chance on me. I still remember trekking to Kirkland Hall freshman Fall for your Neuro 80 office hours and patching bipolar cells together for Neuro 115. I had no clue then that you would become my closest mentor for the next couple of years, nor that you would shape so fundamentally how I think and do science. Thank you for believing in me, for everything you have taught me, and for all your encouragement and support along the way. I feel so lucky to have been mentored by you.

My heartfelt gratitude to my colleagues, Wen, Nathan, Maxwell, and Khalila. Wen, thank you for all your words of encouragement through the years and for teaching me IHC. Nathan, thank you for welcoming me to your project and teaching me head-fixed behavior. Maxwell, I would not have survived my first set of stereotax surgeries without you stepping in for me on the earbars. Khalila, I look forward to many more AnyDesk jumpscare in the near future as more analysis gets underway. Thank you all for being such generous colleagues and good friends. We have done exciting science together and I have loved every bit of it.

To my blockmates Max, Tyler, Jeffrey, Alex, and Corwin, you are the most inspiring group of people I have had the pleasure of calling my best friends. You motivate me every single day, you make me laugh until it hurts, and I know that even at the most difficult moments we are never more than a phone call away. I am so lucky to have done Harvard with you. I look forward to our lifetime of friendship.

To my family, who are the foundation of everything. Mom and Dad, I am so lucky to be your son. You have supported every dream I have ever had without hesitation, and the fact that I get to pursue science freely is because you have built a nest of safety, stability and love through years and years of hard work and devoted parenthood. I hope you know how grateful I am for the life you have made possible. Philip, watching you grow has been one of my greatest joys. You inspire me with who you are and who you are becoming, and I look forward to the day we get to take on the world together. Joel, thank you for being my mentor, my friend, and my advocate. You have believed in me with a generosity I am not sure I deserve, and that belief has carried me further than you know. Having you in my corner has meant everything

And to Dr. Doug Melton and Dr. Steven Hyman, thank you for showing me what it looks like to dedicate a life to science with vision and purpose. I aspire to be the scientists that you are.

List of Contributions

The research presented in this thesis was conducted in the laboratory of Professor Bernardo Sabatini at Harvard Medical School. I joined the Sabatini Lab as a freshman and have worked in the lab across four years, supported by funding from the Kempner Institute (KRANIUM and KURE), the Herchel Smith Undergraduate Research Fellowship, the Yun Family Fellows for Revolutionary Thinking, the TIME Fellowship, and the Sabatini Lab. I have received course credit via NEURO99 for Spring (2025, 2026) and Fall (2026).

The project was conceived and designed jointly by Dr. Kevin Mastro and myself. All resources, animals, and workspaces were provided by the Sabatini Lab.

The full aging behavioral cohort for the two-armed bandit task was run jointly by myself, Wen Chun Lee, Khalila Lord-Arond, and Maxwell Kounga. The two-armed bandit task pipeline was developed by Dr. Kevin Mastro, who trained me in its use. All behavioral analysis and computational modeling were performed by myself with guidance and support from Dr. Kevin Mastro. The recursively formulated logistic regression model was originally developed by Dr. Celia Beron and adapted here for application to the aging cohort.

Fluorescence lifetime photometry recordings in freely moving mice were conducted jointly by myself and Dr. Kevin Mastro. Head-fixed fiber photometry recordings were conducted jointly by myself and Nathan Mallet. The FLIPR system and associated recording approaches were developed by Dr. Bart Lodder in the Sabatini Lab.

Progressive ratio behavioral experiments were designed and conducted by Dr. Kevin Mastro and myself.

The single-nucleus RNA sequencing dataset analyzed in Chapter 5 is a publicly available dataset from Linker et al. (2025), Nature Communications. All reanalysis of this dataset was performed independently.

All figures and written work in this thesis are my own unless otherwise stated.

Table of Contents

ACKNOWLEDGEMENTS	3
LIST OF CONTRIBUTIONS	4
ABSTRACT	7
CHAPTER I: INTRODUCTION	8
<i>1.1 Plasticity as the Foundation of Learning Across the Lifespan</i>	8
<i>1.2 Dopamine as a Neuromodulatory Gate for Plasticity and Learning</i>	9
<i>1.3 The Nucleus Accumbens as a Circuit Substrate for Dopamine-Dependent Plasticity</i>	13
<i>1.4 Aging-Related Alterations in Dopamine and Consequences for Learning</i>	16
<i>1.5 Central Knowledge Gap and Experimental Objectives</i>	18
CHAPTER II: MATERIALS AND METHODS	20
<i>2.1 Animals</i>	20
<i>2.2 Surgeries</i>	21
<i>2.3 Handling, Water Restriction, and Food Restriction</i>	23
<i>2.4 Viruses</i>	24
<i>2.5 Pharmacology</i>	25
<i>2.6 FLIPR Lifetime Photometry</i>	25
<i>2.7 Behavioral Setup</i>	27
<i>2.8 Recursive Formulation of Logistic Regression Model</i>	31
<i>2.9 snRNA-seq Analysis</i>	32
<i>2.10 Histology and Imaging</i>	35
<i>2.11 Ethics</i>	35

CHAPTER III: RESULTS	36
3.1 Acquisition of the Two-Armed Bandit Task Across the Lifespan	36
3.2 Behavior During Probabilistic Execution of the Two-Armed Bandit Task	39
3.3 Interpretable modelling of probabilistic updating with a recursively formulated logistic regression (RFLR) model.....	42
3.4 Dopamine Transients Encode Reward History in the Two-Armed Bandit Task	45
3.5 Aging Reduces Tonic Dopamine Baseline in the Nucleus Accumbens	48
3.6 Tonic Dopamine Baseline Does Not Covary with Motivational State	51
3.7 Aging drives coordinated transcriptional remodeling across striatal cell types.....	54
CHAPTER IV: DISCUSSION	57
4.1 Selective vulnerability of tonic dopamine signaling in aging	57
4.2 Exploratory Molecular Evidence at Dopaminergic Terminals.....	58
4.3 Functional Implications of Reduced Tonic Dopamine Baseline	61
4.4 Behavioral Experience May Preserve Reinforcement-Related Circuit Function Across Aging	63
4.5 Conclusions and Implications	64
4.6 Limitations of the Study	64
REFERENCES	66
SUPPLEMENTARY	72

Abstract

The ability to flexibly update decisions based on changing outcomes is fundamental to adaptive behavior. Whether this capacity deteriorates during normative aging, and what circuit mechanism underlies any such change, remains incompletely understood. Using a two-arm bandit probabilistic reversal task in freely moving mice, we show that aging progressively impairs reinforcement-guided updating across the lifespan. Older mice accumulated more unrewarded outcomes and were slower to re-establish preferences following contingency reversals, despite intact task acquisition and maintained motivational effort. Computational modeling revealed that aging selectively reduced sensitivity to recent reward outcomes while prolonging the persistence of prior evidence, together producing a signature of rigid decision making. Tracking dopamine dynamics in the nucleus accumbens across the lifespan using fluorescence lifetime photometry, we found that tonic dopamine baseline declined progressively with age while phasic transient amplitudes were preserved, compressing the absolute dynamic range of reward signaling without altering its relative structure. Reanalysis of published single-nucleus RNA sequencing data from young and aged striatum revealed coordinated transcriptional changes across cell types, including reactive microglial activation, loss of astrocytic glutamate transporters, and bioenergetic stress, consistent with degraded fidelity of reinforcement signals and impaired translation into behavioral updating. Together, these findings establish a normative trajectory of cognitive aging and correlatively link progressive impairment in outcome-guided decision making to the erosion of dopaminergic tone within a broadly remodeled striatal environment.

I. Introduction

A. Plasticity as the Foundation of Learning Across the Lifespan

As people age, adapting to a changing world becomes increasingly difficult. Older individuals take longer to learn new routines, struggle when rules shift unexpectedly, and persist with strategies that stopped working long ago. These difficulties are familiar enough to feel inevitable, yet their neurobiological basis remain incompletely understood.

At their core, they reflect a decline in cognitive flexibility, the capacity to revise behavior as environmental contingencies change. Cognitive flexibility emerges from the coordinated activity of multiple prefrontal and limbic circuits, whose capacity to adapt in response to experience is grounded in synaptic plasticity, the conversion of transient patterns of neural activity into persistent modifications of connection strength (Citri & Malenka, 2008; Pan & Monje, 2020; Oberman & Pascual-Leone, 2013). Through such modifications, organisms form predictive relationships between actions and outcomes, develop expectations about value, and revise behavioral policies when contingencies change (Whyte et al., 2019). Informative outcomes selectively update synaptic connections that supported the preceding behavior, gradually shifting the probability of future choices and enabling the continuous refinement of choice across experience.

This capacity has been shown to weaken with age. Older adults display characteristic impairments in reinforcement learning, reversal learning, and tasks requiring continuous updating of behavioral policies (Breton et al., 2015; Oberman & Pascual-Leone, 2013; Voglewede et al., 2019; Marzola et al., 2023). Aged rodents display parallel deficits, including

slower adaptation to changing reward contingencies and reduced capacity to shift strategies (Schoenbaum et al., 2002; Zhang et al., 2022; Tait et al., 2013).

At the cellular level, aging reduces the induction and maintenance of long-term potentiation, increases susceptibility to long-term depression, and disrupts the transcriptional programs required for late-phase plasticity (Norris et al., 1996; Foster & Kumar, 2007; Ducote et al., 2024; Boric et al., 2008; Wang, 2008; Chepkova et al., 2015). The resulting system remains capable of learning but becomes progressively biased toward preserving established representations over updating them in response to new experience (Pauwels et al., 2018; James et al., 2024).

Synaptic modification, however, is not governed by local activity-dependent processes alone. Neuromodulatory systems impose an additional layer of regulation (Speranza et al., 2021).

Dopamine, in particular, signals whether outcomes were better or worse than expected, gates synaptic eligibility for modification, and specifies the direction and magnitude of change (Pignatelli & Bonci, 2015; Vidal-Gadea & Pierce-Shimomura, 2012). The integrity of this signal is a prerequisite for flexible behavioral updating, and its degradation with age may represent an alternate route by which neural circuits become resistant to modification.

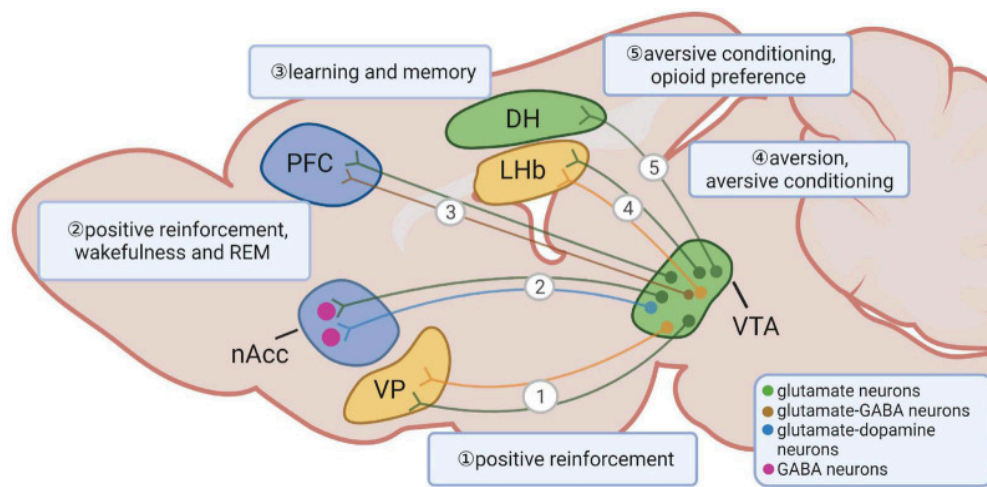
Characterizing how dopaminergic signaling changes with age, and how those changes shape behavior, therefore offers a tractable approach to understanding cognitive decline.

B. Dopamine as a Neuromodulatory Gate for Plasticity and Learning

Resolving dopamine's role in gating synaptic plasticity requires considering (1) the anatomical organization of dopaminergic projections, (2) the computational signals those projections

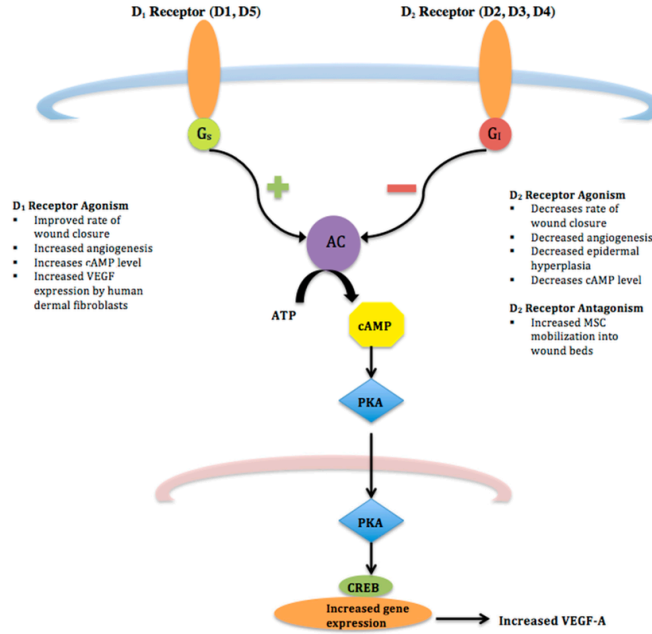
encode, and (3) the mechanisms through which those signals produce lasting synaptic modifications.

Dopamine neurons in the ventral tegmental area (VTA) and substantia nigra pars compacta (SNc) project broadly across the striatum and prefrontal cortex (see Schematic 1), positioning them primarily to modulate circuits implicated in action selection, value estimation, and behavioral adaptation (Menegas et al., 2015; Trutti et al., 2019; Cai & Tong, 2022; Haber, 2011).



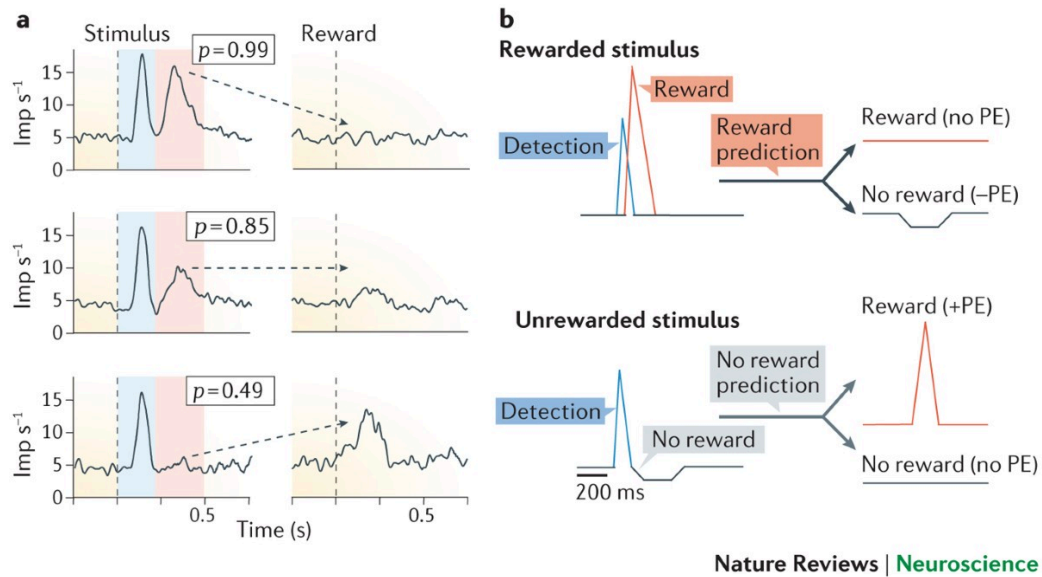
Schematic 1: Long-range projections of VTA glutamate and dopamine neurons and their major targets (Cai & Tong, 2022).

The functional influence of these projections is mediated through two families of G protein-coupled receptors with opposing downstream consequences. D1-class receptors promote intracellular cascades that facilitate synaptic strengthening, while D2-class receptors suppress those cascades and favor weakening, a distinction whose cellular and circuit-level elaboration is taken up in **Section C** (Gerfen et al., 1990; Zhuang et al., 2021; Teng et al., 2022). The direction of plasticity thus depends on which receptor population is preferentially engaged, and this distinction is determined by the content of the dopamine signal.



Schematic 2: Divergent G-protein signaling cascades downstream of D1 and D2 receptors (Vaughn et al., 2018)

A defining computational property of midbrain dopamine neurons is their encoding of reward prediction errors (RPE), expressed as transient increases in firing when outcomes exceed expectations and brief pauses when outcomes fall short (Schultz, 2016; Rescorla & Wagner, 1972). This bidirectional signal directly parallels the error term in temporal difference learning, providing a moment-by-moment estimate of how current outcomes differ from predictions (Dayan & Daw, 2008; Katahira, 2015). In vivo, these prediction errors are expressed as fast phasic fluctuations in extracellular dopamine concentration, with positive errors elevating dopamine above baseline and negative errors suppressing it below, thereby engaging D1 and D2 receptor populations (featuring low and high affinities respectively, such that dopamine rises preferentially activate D1 while dopamine dips disengage D2) and driving plasticity in opposing directions at recently active synapses (Romero Pinto & Uchida, 2025).



Schematic 3: Sustained value coding in dopamine prediction-error signals (Schultz, 2016).

The magnitude of this plasticity, however, is not determined by phasic transients alone. Phasic fluctuations are superimposed on a tonic dopamine baseline that varies slowly relative to discrete signaling events (Floresco et al., 2003; Grace, 1991). This baseline sets the gain on signaling, determining the dynamic range within which phasics are expressed. One might therefore expect that an equivalent transient would reach a higher peak and produce greater synaptic modification when arising from an elevated tonic baseline than a lowered one (Romero Pinto & Uchida, 2025). Tonic and phasic dopamine thus act in parallel, with tonic baseline gating the conditions under which prediction errors are transduced into lasting synaptic change.

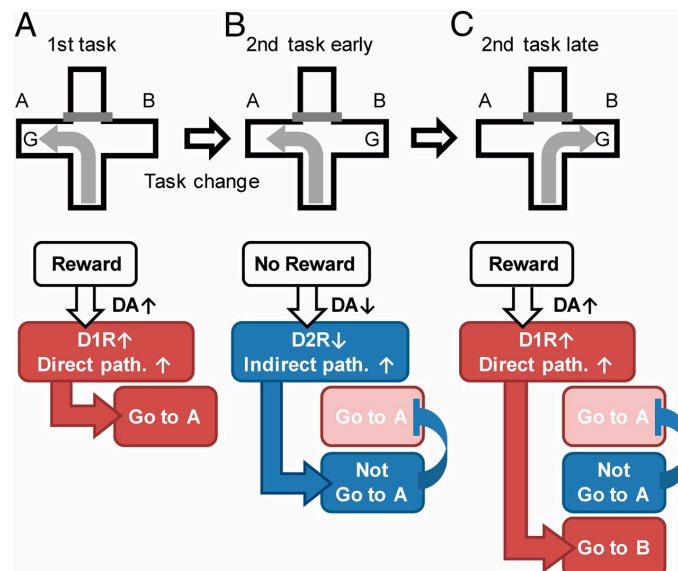
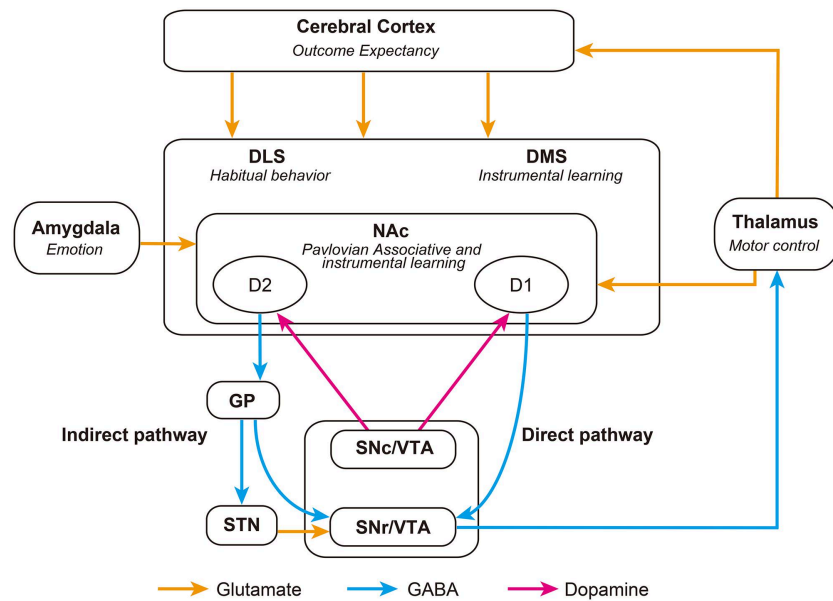
C. The Nucleus Accumbens as a Circuit Substrate for Dopamine-Dependent Plasticity

Dopamine-dependent plasticity does not operate uniformly across the brain. The nucleus accumbens (NAc) represents a site of particular relevance, occupying a central position in the basal ganglia networks that support reward-guided behavior.

As the ventral extension of the striatum, it forms part of cortico-basal ganglia loops linking prefrontal, limbic, and midbrain regions (Xu et al., 2024; Klawonn & Malenka, 2019; Yin et al., 2008). The mPFC and OFC send inputs representing action plans, task rules, and value estimates (Hikida et al., 2020; Lai et al., 2022), the hippocampus contributes contextual information, and the basolateral amygdala conveys emotional and motivational significance of cues and outcomes (Ibrahim et al., 2024; Klawonn & Malenka, 2019). Arousal and salience signals arrive from thalamic and insular regions (Xu et al., 2024) and dense dopaminergic innervation from the VTA delivers moment-to-moment prediction error signals onto this convergent input landscape (Castro & Bruchas, 2019). This arrangement positions the NAc as a site where information about what the animal did, what it expected, and what actually happened is jointly available to be encoded into synaptic weights.

The principal neurons of the NAc are spiny projection neurons (SPNs), which comprise ~90% of the local neuronal population and serve as the cellular substrates through which dopamine prediction error signals are transformed into changes in synaptic strength (Kravitz & Kreitzer, 2012; Shen et al., 2008). SPNs are divided into two populations based on dopamine receptor expression, each forming one arm of the bidirectional plasticity mechanism outlined in **Section**

B. D1-expressing striatomesencephalic SPNs project along the direct pathway and facilitate selected actions, while D2-expressing striatopallidal SPNs project along the indirect pathway and suppress competing or maladaptive ones (Gerfen et al., 1990). Though not absolute, this distinction captures the NAc's fundamental architecture of parallel output channels through which dopamine prediction errors selectively promote or inhibit.

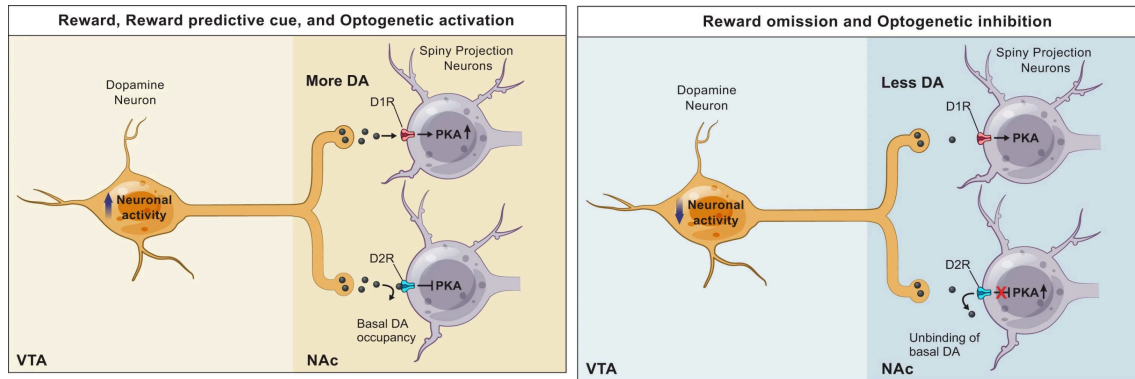


Schematic 5: Representations of striatal-associated decision-making neurocircuitry
(Macpherson et al., 2014; Yawata et al., 2012)

At the cellular level, this directionality is implemented through the opposing consequences of D1 and D2 receptor engagement. Transient increases in dopamine, corresponding to positive prediction errors, preferentially engage D1 receptors on striatomesencephalic SPNs. D1 receptors are positively coupled to adenylyl cyclase, elevating intracellular cyclic AMP and increasing PKA activity, which disinhibits CaMKII at recently active spines and drives the structural and synaptic changes underlying long-term potentiation at excitatory corticostriatal inputs (Shen et al., 2008; Gerfen et al., 1990; Yamaguchi et al., 2022). Conversely, transient dips below baseline, corresponding to negative prediction errors, reduce D2 receptor occupancy on striatopallidal SPNs. D2 receptors are negatively coupled to adenylyl cyclase, and their disengagement elevates PKA activity in these neurons, shifting the intracellular balance toward long-term depression at active synapses, though the precise molecular steps linking D2 disengagement to depression remain less fully characterized than D1 (Shen et al., 2008; Gerfen et al., 1990; Yamaguchi et al., 2022).

In vivo imaging combining dopamine sensors with PKA activity reporters confirms that this model operates during natural behavior, with dopamine elevations driving PKA activation in D1-SPNs and dopamine dips driving PKA activation in D2-SPNs (Lee et al., 2021). Iterated over many trials, this bidirectional modulation selectively strengthens NAc inputs that predict reward and weakens those that do not, producing the experience-dependent tuning of action value representations that underlies flexible reward-guided behavior.

In this way, the NAc functions as the circuit-level substrate where dopaminergic prediction errors are converted into lasting synaptic changes, and through which the incremental refinement of behavioral choice is achieved.



Schematic 6: Opposing dopamine control of PKA signaling in NAc D1 and D2 SPNs (Lee et al., 2021)

D. Aging-Related Alterations in Dopamine and Consequences for Learning

Dopaminergic function has been shown to decline across aging, though the form of this decline varies and no single mechanistic account explains all observations (Morrison & Hof, 1997).

Several studies report steady reductions in striatal dopamine content across age, with dopamine and its associated metabolites falling to a fraction of their peak levels from early adulthood into late life (Goldman-Rakic and Brown 1981; Kish et al., 1992). These reductions frequently occur even when presynaptic terminals remain structurally present and when markers of dopamine synthesis, vesicular packaging, and reuptake show relatively modest or region-specific change (Kish et al., 1992). This finding suggests that aging can limit the amount of transmitter available for release while leaving much of the underlying anatomical scaffold intact.

Other observations emphasize alterations in postsynaptic sensitivity. In several rodent models, total striatal dopamine concentrations remain stable well into advanced age, yet the availability of D1 and D2 receptors declines, and intracellular signaling responses become weaker (Karrer et al., 2017). Various human imaging studies echo this pattern, but also reveal substantial individual variability in the extent and spatial distribution of receptor and transporter reductions (Dang et al., 2017). These data would suggest that aging affects the ability of postsynaptic neurons to detect and respond to dopamine even when presynaptic stores of transmitter are relatively preserved.

Age-related vulnerability in midbrain dopamine neurons adds another dimension. Structural imaging studies report reduced integrity within the substantia nigra in older adults, and histological work confirms modest neuronal loss over the lifespan (Fearnley and Lees, 1991).

Juxtaposed, these findings suggest that dopaminergic signaling declines with age, but through mechanisms that are neither uniform nor reducible to a single biological source.

These biological shifts are nonetheless associated with consistent functional consequences in the neural circuits through which dopamine shapes behavior. Functional neuroimaging studies show that older human adults often exhibit reduced ventral striatal engagement during reward anticipation and altered timing of reward responses in both striatum and midbrain (Eppinger et al., 2013; Dreher et al., 2008). Aging rodents display comparable reductions in dopamine-dependent responses to reward-predictive cues (Tryon et al., 2020; Lei et al., 2025).

Behaviorally, these neural changes parallel well-characterized deficits in reinforcement learning. Older adults and aged rodents learn more slowly from positive outcomes, update value estimates less effectively, and struggle to flexibly adjust choices when reward contingencies shift (Cools et

al., 2009). In dynamic or probabilistic tasks, many older individuals show reduced trial-by-trial sensitivity to feedback and a greater tendency toward perseverative strategies (Rutledge et al., 2009).

Pharmacological studies strengthen the link. In older adults, enhancing dopaminergic transmission with levodopa can increase learning rates and restore more complete reward prediction error signals in the nucleus accumbens (Chowdhury et al., 2013). The degree of behavioral improvement varies across individuals, but in those who benefit, dopamine enhancement reinstates the canonical relationship between expected value, reward outcome, and striatal prediction-error responses (Chowdhury et al., 2013; Moustafa, 2011). These findings offer direct evidence that dopaminergic insufficiency contributes to impaired reward-guided learning in aging.

E. Central Knowledge Gap: How Does Aging Alter Dopamine-Dependent Plasticity in the Nucleus Accumbens?

Although aging is known to disrupt dopamine signaling and impair reinforcement learning, the mechanisms that link these changes remain poorly understood. Dopamine shapes the plasticity rules through which the NAc integrates reward history, yet it is not known how aging alters the mapping from dopamine signals to synaptic plasticity, or from plasticity to behavior. This thesis addresses this gap by employing a behavioral assay of cognitive flexibility and *in vivo* fluorescence-lifetime measurements of dopamine. The two-armed bandit task (2-ABT) provides a readout of trial-by-trial reward learning, outcome sensitivity, and adaptability when contingencies change (Mastro et. al, 2024). Lifetime dopamine photometry (FLIP-R) enables direct quantification of extracellular dopamine fluctuations independent of intensity-based

distortions, following recent advances that demonstrate stable lifetime-based dopamine readouts in behaving animals (Lodder et al., 2025). By tracking these signals across the mouse lifespan, this work seeks to investigate the potential role of age-dependent neuromodulator dynamics in striatal computation and behavior.

Central hypothesis. Aging reduces dopamine in the NAc, impairing the fidelity with which reward outcomes drive behavioral updating. As a consequence, decision-making becomes less sensitive to recent outcomes, reinforcement learning slows, and behavioral flexibility declines.

Aim 1. Characterize how cognitive flexibility changes across the lifespan: Mice at adolescent, adult, and aged timepoints will perform a two-arm bandit task (2-ABT) requiring continuous updating of port identity and rapid adaptation when reward contingencies reverse. Computational modeling previously developed in the lab will be utilized to quantify behavioral shifts across age (Beron et al., 2022). I expect that aging will slow reward-based updating, reduce flexibility after reversals, and increase reliance on habitual or perseverative policies. This aim establishes the behavioral phenotype that subsequent physiological and molecular analyses seek to explain.

Aim 2. Quantify lifespan changes in tonic and phasic dopamine signals using fluorescence-lifetime dopamine photometry: To measure baseline dopaminergic tone and reward-evoked dopamine fluctuations in the nucleus accumbens, I will use fluorescence lifetime dopamine photometry (FLIP-R). Lifetime-based readouts provide a stable and quantitative measure of neuromodulatory dynamics across age. I predict that aged animals will exhibit reduced dopamine levels and thus diminished reward/information-evoked phasic dopamine

responses. These changes are expected to weaken the effective teaching signals underlying updating.

Aim 3. Identify age-dependent structural and molecular correlates of dopamine decline via transcriptomic profiling: To determine mechanisms underlying age-related dopamine reductions, I will perform transcriptomic profiling via re-analysis of a public mouse striatum snRNA-seq dataset at young and aged timepoints. I will seek to identify age-dependent transcriptional changes. I predict that aging will be associated with structural or molecular adaptations with transcriptional basis in dopamine-relevant circuits.

II. Materials and Methods

All experimental procedures were conducted in accordance with the National Institutes of Health guidelines and were approved by the Harvard Medical School Institutional Animal Care and Use Committee (IACUC).

Animals

Only Male mice were used across experiments. Wild-type C57BL/6J mice (n=19) used for lifetime dopamine photometry and FR/PR experiments were obtained from the National Institute on Aging (NIA) Aged Rodent Colonies, which maintain C57BL/6J mice across a broad age range (1–30 months). D1-Cre C57BL/6J mice (n=6) used for dopamine and D1-SPN/D2-SPN PKA (not included in thesis) experiments were obtained from The Jackson Laboratory.

Wild-type C57BL/6J mice (n= 22) used for head-fixed dopamine photometry were obtained from The Jackson Laboratory. These animals formed the primary cohort for experiments examining age-dependent dopamine signaling and behavior.

Wild-type C57BL/6J mice (n=80 total; n=76 progressed to 80/20) used for 2-arm bandit behavioral experiments were obtained from The Jackson Laboratory.

Mice were housed in a temperature- and humidity-controlled vivarium on a 12 h light/dark cycle with ad libitum access to food. Behavioral experiments were conducted during the dark phase of the light cycle. At the time of surgery, animals spanned early adulthood through advanced aging, enabling cross-sectional and longitudinal analyses across the lifespan. All animals were group-housed following surgical implantation, singly housed following the onset of behavioral experiments.

Surgeries

Mice were anesthetized with isoflurane (induction at 3–4%, maintenance at 1–2%) and placed in a stereotaxic frame. Body temperature was maintained using a closed-loop heating pad, and depth of anesthesia was continuously monitored. Following scalp incision and skull exposure, bregma and lambda were aligned in the dorsoventral plane. Craniotomies were drilled above target regions using stereotaxic coordinates referenced to bregma.

For nucleus accumbens core (NAcc) targeting, viral injections were performed at the following coordinates: anterior–posterior (A/P) +1.3 mm, medial–lateral (M/L) $\pm$ 1.3 mm, dorsal–ventral (D/V) –4.1 mm. Injection pipettes were lowered beyond the target depth (–4.15 mm) and retracted to the final depth prior to infusion to minimize backflow. Viruses were delivered using

pulled glass pipettes at a rate of 80 nL/min, with total volumes of 300 nL per site unless otherwise specified. Following infusion, pipettes were left in place for an additional 5 min before slow withdrawal.

Following viral injection, the scalp was then retracted to fully expose the skull, which was cleaned, lightly etched, and dried to promote adhesion. Skin edges were secured to the skull perimeter using tissue adhesive (Vetbond) to prevent soft tissue encroachment. Pre-cut optical fibers (200/230 μm core/cladding, NA 0.37, 4.5 mm length; Doric Lenses) were mounted in a stereotaxic implant holder and lowered slowly to a position 100 μm dorsal (-4.0 mm) to the injection site. Fibers were positioned to avoid contact with the injection tract and allowed to settle prior to fixation.

Fiber implants were secured to the skull using a thin layer of cyanoacrylate adhesive (Loctite) applied at low temperature to prevent premature curing and carefully distributed around the fiber–skull interface while avoiding the fiber tip. Adhesive was rapidly polymerized using an accelerator (ZipKicker), after which the implant holder was gently removed. A stainless steel headbar was positioned posterior to the fiber implant and affixed using the same cyanoacrylate-based procedure.

To reinforce implant stability, a dental cement (Metabond; powder:base:catalyst at a 1:3:1 ratio) was mixed immediately prior to use and applied in three successive layers around the base of the fiber implant, headbar, and exposed skull. Each layer was allowed to partially cure before application of the next, forming a rigid, sealed headcap. Implant integrity and alignment were verified before recovery.

Postoperatively, mice received analgesia and were allowed to recover for a minimum of 2–3 weeks prior to behavioral testing and photometry recordings to ensure stable viral expression and full recovery.

Handling, Water Restriction, and Food Restriction

Prior to 2-arm bandit behavioral training, mice were gradually water restricted and handled daily to acclimate them to the experimenter and behavioral procedures. Water access was limited to approximately 1–3 mL per mouse per day, and body weight was monitored daily to maintain animals at no less than 80% of their baseline weight. Mice were handled for 5 consecutive days before the start of behavioral experiments. Tail handling was avoided, and mice were instead transferred using gentle cupping or tunnel handling. Water restriction and handling procedures were maintained throughout behavioral training and testing.

For FR/PR experiments requiring food-motivated responding, mice were gradually food restricted and handled daily to acclimate them to experimental procedures. Food access was limited to approximately 1–3 g of standard chow per mouse per day, and body weight was monitored daily to maintain animals at no less than 80% of their baseline weight. The precise amount of food provided was adjusted as needed to maintain stable body weight within this range. For experiments involving fluorescence lifetime photometry and related physiological measurements, food restriction protocols were modified as necessary to ensure stable physiological conditions and consistent lifetime signal measurements while maintaining animal health and weight above the minimum threshold. Food restriction was maintained throughout behavioral testing.

Viruses

Recombinant adeno-associated viruses (AAVs) were used to deliver the sensors that allowed for monitoring of extracellular dopamine dynamics and downstream intracellular signaling. Viral vectors were delivered either in a Cre-independent manner or using cell-type–restricted enhancer or Cre-dependent FLEX configurations as indicated. AAVs were obtained from commercial vector cores or in-house sources. All viral aliquots were stored at -80°C upon arrival and thawed on ice immediately prior to use.

Extracellular dopamine dynamics were measured using genetically encoded dopamine sensors, including red-shifted dLight variants (AAV-hSyn-dLight[chromson]) and GRAB-DA3.0 (AAV PHP.eb-hSyn-DA3.0). For dLight experiments targeting the nucleus accumbens (NAc), viruses were used at an initial titer of $\sim 1.1 \times 10^{13}$ genomic copies (gc)/mL and diluted to a working concentration of $\sim 7 \times 10^{12}$ gc/mL. Typical injection volumes were 300 nL per site, infused at 50 nL/min. For GRAB-DA3.0 experiments, working titers were $\sim 1.3 \times 10^{13}$ gc/mL with injection volumes of 250–300 nL per site.

Intracellular PKA signaling was measured using fluorescence lifetime–based reporters (FLIM-AKAR). Both enhancer-driven and Cre-dependent FLEX FLIM-AKAR constructs were employed to achieve cell-type specificity within the striatum. Enhancer-based FLIM-AKAR variants selective for D1- or D2-expressing striatal projection neurons were injected at titers ranging from $\sim 5.0 \times 10^{12}$ to 9.2×10^{13} gc/mL, with injection volumes of 150–300 nL per site. In D1-Cre mice, FLEX FLIM-AKAR was additionally used to selectively target D1 receptor–expressing neurons. In several experiments, dLight and FLIM-AKAR viruses were

co-injected to enable simultaneous measurement of dopamine release and downstream intracellular signaling.

Across all experiments, injection volumes ranged from 150–300 nL per site and infusion rates were carefully controlled to minimize tissue damage and backflow. The FLEX FLIM-AKAR reporter used in this study is available through Addgene (AAV-FLEX-FLIM-AKART391A, #60446). Dopamine sensor constructs (dLight variants and GRAB-DA3.0) were obtained through the Sabatini laboratory or UNC Neurotools and are available upon request.

Pharmacology

The D1 dopamine receptor antagonist SCH23390 (Tocris Bioscience, #0925) was dissolved in sterile 0.9% saline and administered via intraperitoneal injection at a dose of 10 mg/kg (prepared at 1 mg/mL). Injections were performed using a fine-gauge syringe (BD 324911, BD Biosciences) and were typically completed within 5–20 s of the onset of scruffing to minimize handling-related variability. Photometry recordings were conducted following drug administration as specified in the corresponding experimental protocols.

FLIPR Lifetime Photometry

Fluorescence lifetime photometry was performed using a fluorescence lifetime photometry at high temporal resolution (FLIPR) system set up in the Sabatini laboratory based on published designs (Lodder, 2025). Optical excitation light from a pulsed 473 nm laser (50 MHz repetition rate) was delivered through an optical patch cord to an implanted multimode optical fiber stub. Emitted fluorescence was collected through the same fiber, spectrally separated from excitation light, and detected by a photomultiplier tube (PMT). Fluorescence lifetime was computed in real time using frequency-domain analog phase comparison between the PMT emission signal and

the laser reference (Lodder, 2025), yielding continuous measurements of fluorescence lifetime in nanoseconds (ns) alongside fluorescence intensity.

The analog processing unit low-pass filtered the laser reference and PMT signals to isolate the 50 MHz components and computed phase-dependent intermediate-frequency voltages using four phase-shifted reference channels ($0, 0.5\pi, \pi, 1.5\pi$). These intermediate-frequency signals and the DC intensity component were digitized using a data acquisition (DAQ) device and converted to lifetime and phasor components using custom MATLAB acquisition software (Lodder, 2025).

Calibration

Calibration of the FLIPR system was performed using the fluorescence lifetime standard Coumarin 6 dissolved in ethanol, which has a known fluorescence lifetime of approximately 2.40 ns. Calibration measurements were acquired through optical fibers of identical configuration to those used in vivo. Phase alignment, voltage offset correction, and system response calibration were performed as described previously (Lodder, 2025). Calibration was performed prior to experimental recordings to ensure accurate and stable lifetime measurements across sessions.

Quantification

Tonic baseline dopamine was quantified as the modal fluorescence lifetime value within each session. Phasic dopamine events were quantified by detecting local maxima in the fluorescence lifetime trace using MATLAB findpeaks with a minimum peak prominence threshold of 0.075 ns and extracting the resulting peak times and peak values. Peak detection outputs were visually inspected to confirm that detected peaks corresponded to transient lifetime excursions rather than

noise. Relative phasic magnitude was computed as peak-minus-baseline and peak-to-baseline ratio.

Behavioral Setup

Fixed Ratio/ Progressive Ratio Sugar Pellet Task

In fixed ratio (FR) sessions, mice were required to perform a fixed number of responses to obtain a reward. Task parameters were adjusted to probe effort-independent dopamine signaling and establish baseline response–outcome contingencies.

In progressive ratio (PR) sessions, the number of responses required for reward delivery increased systematically across trials. Breakpoints were defined as the final completed ratio prior to cessation of responding and were used as a behavioral index of motivational state and effort valuation.

Mice were trained on operant tasks using a staged protocol designed to progressively shape instrumental responding and assess effort-based motivation. Prior to training, animals underwent mild food restriction to motivate task engagement while maintaining health, and were maintained at approximately 80% of free-feeding body weight with daily monitoring. Training proceeded sequentially through free-feeding exposure, fixed-ratio schedules, and finally a progressive ratio schedule, with advancement contingent on behavioral criterion rather than elapsed time.

Training began with free-feeding exposure to familiarize animals with the reward pellets and feeding device. Mice were then trained on a fixed-ratio 1 (FR1) schedule, in which each response resulted in delivery of a single sugar pellet. Upon meeting criterion performance (earning 30

pellets within a session), animals were advanced to a fixed-ratio 3 (FR3) schedule, in which three responses were required per reward, again with a criterion of 30 pellets per session.

Daily training sessions during free-feeding, FR1, and FR3 stages were standardized to 3 h in duration. Once animals met FR3 criterion, they were transitioned to a progressive ratio (PR) schedule to assess effort-based motivation. In PR sessions, the response requirement increased systematically after each earned reward, and sessions were not capped by pellet number. PR sessions ran for 6 h in duration.

All operant training and testing was performed using the FED3 (Feeding Experimentation Device version 3), which automatically delivered 20 mg sugar pellets contingent on behavioral responses and recorded trial-by-trial behavioral data. Training schedules were staggered to allow the introduction of new animals while maintaining consistent task structure for previously trained cohorts.

2-Arm Bandit Lick Task

The two-armed bandit task (2-ABT) was implemented using the same behavioral apparatus and task logic as previously described studies, with minor adaptations for aging and photometry experiments. Briefly, mice were trained to perform a center-initiated choice between two side ports whose reward probabilities reversed in an uncued manner across blocks. The task was designed to probe reinforcement learning, behavioral flexibility, and reward history integration.

Behavior was conducted in custom-built operant chambers (4.9" × 6" acrylic) equipped with three nose ports (center, left, right), each monitored by infrared beam-break sensors. Rewards consisted of liquid sucrose solution delivered via solenoid-controlled valves. Behavioral control

and data acquisition were implemented using custom software, and all training and testing occurred during the dark phase of the light cycle under no ambient illumination.

Task Acquisition and Shaping: To facilitate acquisition of the task, mice progressed through a series of structured training stages in which the center-to-side action sequence and decision timing were gradually shaped (Supplementary Table 1). Across stages, key task parameters, including the allowable decision window between center and side pokes, reward probabilities associated with each side port, and the length of blocks over which contingencies remained constant, were adjusted.

In Stage 1, mice were rewarded for pokes into either side port to establish the basic action–reward contingency. In Stage 2, trials required initiation via a center nose poke followed by a side port selection, establishing the center-to-side action sequence. This association was progressively strengthened across Stages 2–5 by reducing the decision window (the maximum time permitted between the center poke and side poke), thereby encouraging rapid, deliberate choices. In Stage 6, block lengths were shortened to promote sensitivity to contingency changes. Advancement through Stages 1–5 required acquisition of 100 rewards within a single session, while advancement from Stage 6 required acquisition of 200 rewards.

Probabilistic Execution Phase: Upon successful completion of the shaping stages, mice advanced to Stage 7, which constituted the execution phase of the two-armed bandit task. In this final stage, one side port was designated as the high-reward option ($p = 0.8$) and the other as the low-reward option ($p = 0.2$). Reward contingencies reversed unpredictably in uncued blocks of variable length (10–30 trials), requiring animals to continuously update action values based on recent reward history.

For initial task execution experiments, mice completed 10 sessions of the probabilistic task at the 80/20 contingency. For retraining experiments, mice completed 15 execution sessions at the same 80/20 contingency without repeating earlier shaping stages. These execution-only sessions were used to assess stable performance, learning dynamics, and age- or manipulation-dependent differences in reinforcement learning.

All behavioral sessions were conducted during the dark phase of the sleep–wake cycle. Only completed trials, defined as trials in which a center poke was followed by a valid side poke, were included in behavioral analyses.

Head-Fixed

For head-fixed recordings, stainless steel head bars were secured to a custom-designed adjustable rigid holder to stabilize the skull while allowing the body to remain supported within a custom cylindrical platform.

Mice performed the two-arm bandit task using two lick ports positioned within reach of the animal. Licks were detected using capacitive sensors, and rewards consisted of water delivery contingent on probabilistic reward schedules identical to those used in the freely moving configuration. Behavioral events and fluorescence intensity signals were recorded simultaneously.

Recursive Formulation of Logistic Regression (RFLR) Model

2-ABT behavior was modeled using the recursive formulation of logistic regression (RFLR), as described in Beron et al., 2022. This model provides a compact and biologically plausible description of probabilistic decision-making by integrating recent choice–outcome history with an explicit bias toward repeating previous actions.

On each trial t , the model computes a latent decision variable ψ_{t+1} representing the log-odds of selecting one port over the alternative:

$$\psi_{t+1} = \alpha c_t + \phi_t$$

where $c(t)$ represents the animal's choice on trial t , encoded as +1 for rightward choices and -1 for leftward choices. The parameter α captures a choice history bias, also referred to as stickiness, which reflects a tendency to repeat the previous action independent of outcome.

The accumulated evidence term ϕ_t is updated recursively according to:

$$\phi_t = \beta(c_{t-1}r_{t-1}) + e^{-1/\tau}\phi_{t-1}$$

where $r(t)$ denotes trial outcome, encoded as +1 for rewarded trials and 0 for unrewarded trials.

The interaction term $c(t)r(t)$ represents signed evidence favoring or opposing the chosen action.

The parameter β determines the strength of evidence integration, and τ controls the timescale over which past evidence decays.

The probability of selecting the right port on the subsequent trial is given by the logistic function:

$$P(\text{choice}_{t+1} = \text{right}) = 1 / (1 + \exp(-\psi_{t+1}))$$

Model parameters α , β , and τ were estimated by maximizing the likelihood of observed trial-by-trial choices. This recursive formulation enables efficient computation by maintaining a single internal state variable that represents exponentially weighted choice–outcome history.

snRNA-seq Analysis

Data acquisition and preprocessing. Single-nucleus RNA sequencing data were obtained from two publicly available datasets: GSE225741 (2-month-old mice, $n = 2$ biological replicates) and GSE226138 (18-month-old mice, $n = 4$ biological replicates), both generated from whole striatum using the 10x Genomics Chromium platform. Raw count matrices were loaded using Scanpy (v1.11.1) via `sc.read_10x_mtx`. All samples were concatenated into a single AnnData object with sample and age group labels retained.

Quality control. Mitochondrial gene content was calculated for each nucleus (mouse mitochondrial genes identified by Mt- prefix). Nuclei were retained if they expressed between 600 and 3,500 genes and had fewer than 5% mitochondrial reads. Genes expressed in fewer than 3 nuclei were excluded. These thresholds were matched to those reported in the original data publications.

Doublet detection. Putative doublets were identified independently per sample using Scrublet (expected doublet rate = 6%), with doublet scoring performed on 20 principal components. Predicted doublets were removed prior to downstream analysis.

Normalization and feature selection. Raw counts were normalized to 10,000 counts per nucleus and log-transformed ($\log_1 p$). Top 6,000 highly variable genes were selected using the Seurat v3 variance-stabilizing transformation with sample as a batch covariate

(sc.pp.highly_variable_genes, flavor='seurat_v3', batch_key='sample'). Expression values were scaled to zero mean and unit variance, clipping at a maximum value of 10.

Dimensionality reduction and batch integration. Principal component analysis (PCA) was performed on the highly variable genes (30 components). To correct for sample-level batch effects while preserving biological variation, Harmony integration was applied to the PCA embedding using harmonypy (v0.2.0), with a diversity penalty $\theta = 2.0$ and a maximum of 30 iterations (batch_key='sample'). A nearest-neighbor graph was constructed on the Harmony-corrected embedding using 15 neighbors and 10 principal components (sc.pp.neighbors). UMAP coordinates were computed for visualization.

Clustering and cell type annotation. Leiden clustering was performed at a resolution of 0.08, yielding 10 clusters. Cell types were assigned manually by inspection of canonical striatal marker gene expression, visualized as dot plots and UMAP feature plots. Marker genes used for annotation included: Drd1, Pdyn (D1-SPN); Drd2, Penk (D2-SPN); Chat, Slc18a3 (cholinergic interneurons); Pvalb, Sst, Npy (GABAergic interneurons); Aldoc, Aqp4, Sox9 (astrocytes); Plp1, Mbp (oligodendrocytes); Pdgfra (OPCs); Cx3cr1, P2ry12 (microglia); Cldn5 (endothelial); Acta2, Rgs5 (mural cells/pericytes). Cluster-level marker genes were also computed using Wilcoxon rank-sum tests (sc.tl.rank_genes_groups) to confirm annotations.

Differential expression analysis. Aging-associated differential expression was assessed within each annotated cell type independently. Nuclei from 18-month-old animals were compared to 2-month-old animals using the Wilcoxon rank-sum test (sc.tl.rank_genes_groups, method='wilcoxon'), applied to log-normalized counts. Cell types with fewer than 10 nuclei per age group were excluded. P-values were corrected for multiple comparisons using the

Benjamini-Hochberg procedure. Genes were considered differentially expressed at adjusted $p < 0.05$ and $|\log_2FC| > 0.25$ (summary analyses) or $|\log_2FC| > 0.5$ (GO enrichment analyses).

Statistics. Comparative analyses between age group within clusters were performed using the Mann-Whitney U test (two-sided). Statistical significance was assessed at $p < 0.05$.

Software. All analyses were performed in Python 3.11 using Scanpy (v1.11.1), AnnData (v0.11.4), harmonypy (v0.2.0), Scrublet, gseapy, NumPy, pandas, SciPy, and Matplotlib.

Reanalysis of Kilfeather et al. (2024) TRAP sequencing data

Translating Ribosome Affinity Purification (TRAP) sequencing data from Kilfeather et al. (2024) were reanalyzed from publicly deposited raw count matrices from young and aged male mice across ventral striatum and midbrain/VTA. Raw count matrices were loaded per sample and merged with associated metadata. Library-size normalization was performed by converting counts to counts per million (CPM) followed by $\log_2$ transformation ($\log_2(\text{CPM} + 1)$). Quality control metrics including library size and number of detected genes were assessed per sample prior to analysis. Principal component analysis was performed on the top 2,000 most variable genes to assess sample clustering by region and age.

Differential expression analysis was performed separately per region. For the midbrain, where $n = 8$ per age group drawn from two cohorts, a per-gene Welch's t-test was applied comparing old versus young log-CPM values, with p-values corrected for multiple comparisons using the Benjamini-Hochberg procedure (adjusted $p < 0.05$, $|\log_2FC| \geq 1.5$). For the ventral striatum and bulk total fractions where $n = 1$ per condition, statistical testing was not performed. Genes were ranked by fold change and thresholded at $|\log_2FC| \geq 1.5$. Gene symbols were mapped from Ensembl IDs using the mygene package. Gene Ontology Biological Process enrichment was

performed on significantly differentially expressed gene lists using gseapy (Enrichr interface, GO_Biological_Process_2023, mouse). Results were filtered to adjusted $p < 0.05$ and visualized as bubble plots. All analyses were performed in Python 3.11 using NumPy, pandas, SciPy, statsmodels, and Matplotlib.

Histology and Imaging

Following behavioral and photometry experiments, mice were deeply anesthetized with isoflurane and transcardially perfused with phosphate-buffered saline (PBS), followed by fixation with 4% paraformaldehyde (PFA). Brains were extracted and post-fixed overnight in 4% PFA at 4 °C, then transferred to 30% sucrose in PBS for cryoprotection until fully equilibrated. Brains were embedded in optimal cutting temperature compound (O.C.T.; Tissue-Tek), rapidly frozen on dry ice, and sectioned coronally at 50 μm using a cryostat. Free-floating sections were mounted using a DAPI-containing antifade mounting medium (Vectashield, H-1200) to preserve fluorescence. Fluorescent reporter expression and optical fiber placements were verified using widefield fluorescence microscopy with a 10 $\times$ objective. Native fluorescence from viral reporters was sufficient for visualization, and no immunohistochemical amplification was performed. Animals with incorrect viral targeting or fiber placement were excluded from further analysis.

Ethics

All experimental procedures were conducted in accordance with the guidelines set by the Institutional Animal Care and Use Committee (IACUC).

III. Results

Acquisition of the two-arm bandit task across the lifespan

To characterize the lifespan trajectory of cognitive flexibility in mice, we trained water-restricted mice spanning six age groups on a two-armed bandit task that requires continuous reinforcement-guided updating. In this task, trials begin with initiation at a center port followed by selection of one of two side ports, resulting in delivery or omission of water reward (Fig. 1a). One port delivers reward at high probability and the other at low probability, with the identity of the high-reward port switching between blocks in an uncued manner. Because the full probabilistic task places significant demands on learning and timing, training followed a structured curriculum of six stages that progressively introduced temporal constraints and probabilistic contingencies (See Materials and Methods). Mice advanced between stages only after meeting predefined performance criteria, ensuring that progression reflected genuine task acquisition rather than passive exposure.

We first asked whether aging altered the efficiency with which mice acquired this curriculum. To quantify this, we examined the number of sessions required to reach the final probabilistic stage across age groups. Acquisition speed declined with age, with older mice requiring progressively more sessions to learn the task structure and reach criterion (Fig. 1b). The effect was largest in the oldest age group, with 18-24 month mice requiring on average approximately 9 more sessions than 0-2 month mice to reach the final probabilistic stage.

To identify what drives slower acquisition in older mice, we compared animals at equivalent stages of learning. Behavior was aligned by normalised learning progress and divided into early,

middle, and late phases. During early acquisition, older mice performed substantially more total trials, accumulating greater numbers of both rewarded and unrewarded outcomes, with the 18-24 month group showing the largest difference (Fig. 1c). Yet despite this greater experience, reward rates were markedly lower in older animals, indicating that the additional trials reflected inefficient learning rather than simply greater task engagement (Fig. 1d). By the middle phase, rewarded trial counts had equalized across age groups, yet older mice continued to accumulate significantly more unrewarded outcomes (Fig. 1c). These differences resolved by the late phase, indicating that all age groups eventually acquired the task structure but that older mice required more experience to do so.

The increased frequency of unrewarded outcomes in aged mice could reflect differences in exploration strategy, such as increased action switching. However, switch probability declined systematically with acquisition and did not differ significantly across age groups at any learning phase (Fig. 1e), indicating that excess unrewarded sampling was not explained by increased exploration.

Alternatively, excess unrewarded outcomes could reflect impaired trial execution rather than learning deficit. Trial duration similarly showed no systematic increase with age after early adulthood, with older age groups performing at comparable speeds despite their elevated error rates (Fig. 1f), suggesting that aged mice were adequately capable of task execution.

Together, these findings demonstrate that aging selectively impairs the efficiency of reinforcement-guided learning, with aged mice requiring substantially more experience to establish the action-outcome contingency.

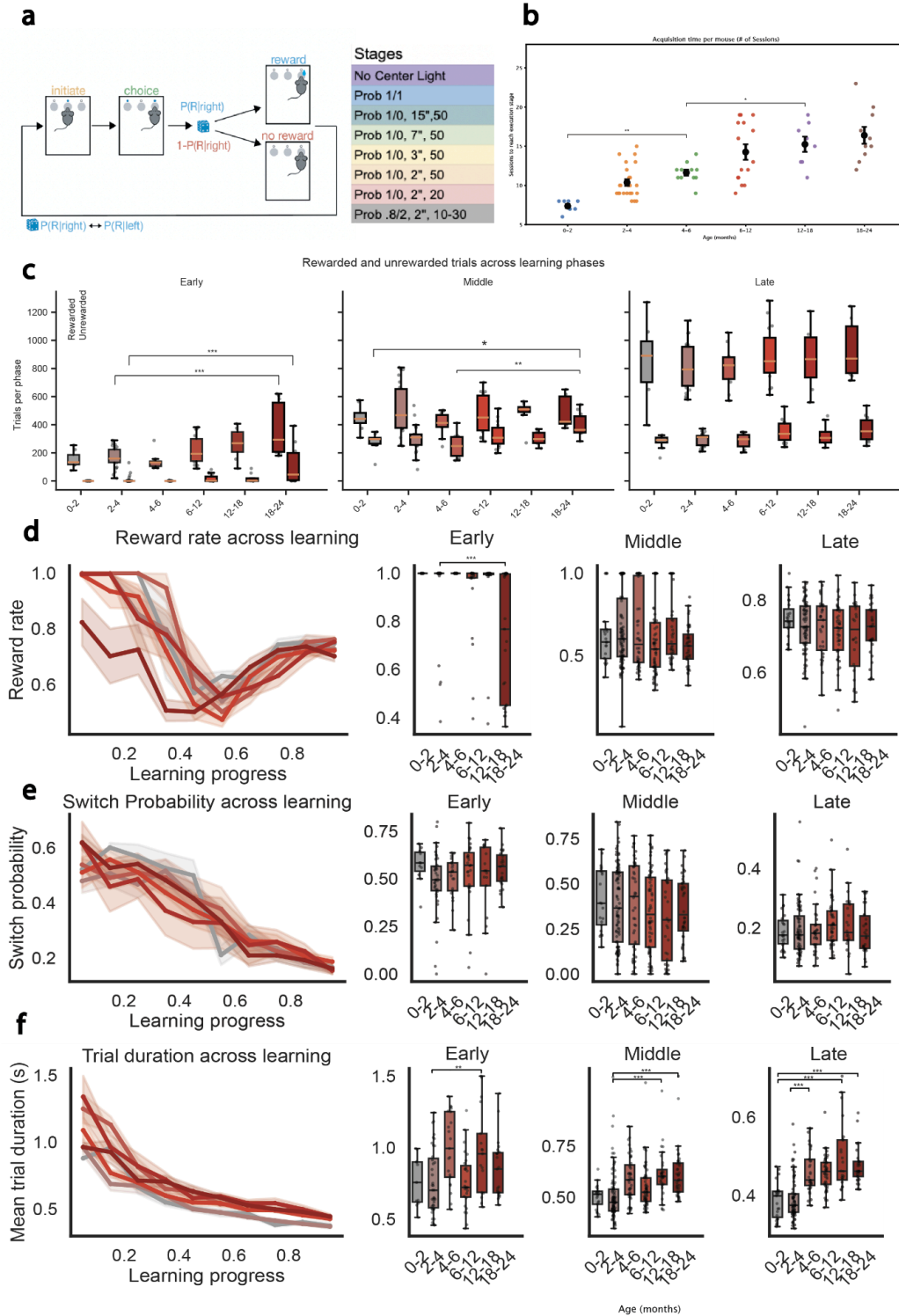


Figure 1. Aging impairs the efficiency of reinforcement-guided task acquisition without disrupting exploration or task engagement.

$n = 8, 27, 9, 17, 8, 11$ mice for age groups 0–2, 2–4, 4–6, 6–12, 12–18, and 18–24 months, respectively. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, Tukey HSD post-hoc following one-way ANOVA unless otherwise stated.

a, Schematic of the two-armed bandit task. Mice initiated trials at a center port and selected one of two side ports, resulting in delivery or omission of water reward. Reward contingencies reversed periodically in an uncued manner. Training progressed through stages of increasing temporal and probabilistic difficulty, culminating in an 80/20 probabilistic contingency with block lengths of 10–30 trials.

b, Sessions required to reach the final probabilistic stage, plotted per mouse (colored dots) as mean $\pm$ SEM. One-way ANOVA: $F(5,74) = 17.06$, $p < 0.001$. Selected pairwise comparisons shown ($***p < 0.001$).

c, Rewarded (left) and unrewarded (right) trial counts across learning phases (early: normalized progress < 0.33 ; middle: $0.33\text{--}0.66$; late: > 0.66), plotted as box plots (median, IQR, whiskers = $1.5 \times \text{IQR}$) with individual animals overlaid. Trial counts are mouse-balanced across phases. Unrewarded trials, early phase: $F(5,70) = 6.37$, $p < 0.001$. Rewarded trials, early phase: $F(5,70) = 7.80$, $p < 0.001$. Unrewarded trials, middle phase: $F(5,70) = 3.62$, $p = 0.006$. Rewarded trials, middle and late phases and unrewarded late phase: not significant or no individual comparisons reached significance. Selected pairwise comparisons shown ($***p < 0.001$).

d, Reward rate across learning. Left, mean $\pm$ SEM per age group plotted against normalized learning progress. Right, reward rate at early, middle, and late phases plotted as in c. Early phase: $F(5,132) = 7.48$, $p < 0.001$. Middle and late phases: not significant ($p = 0.110$ and $p = 0.330$, respectively). Selected pairwise comparisons shown ($***p < 0.001$).

e, Switch probability across learning phases, plotted as in d. One-way ANOVA at each phase: early $F(5,132) = 0.97$, $p = 0.439$; middle $F(5,284) = 0.75$, $p = 0.585$; late $F(5,221) = 0.86$, $p = 0.510$. No significant differences detected at any phase.

f, Trial duration across learning phases, plotted as in d. Early phase: $F(5,132) = 4.74$, $p < 0.001$. Middle phase: $F(5,284) = 9.46$, $p < 0.001$. Late phase: $F(5,221) = 24.95$, $p < 0.001$. Significant differences at early and middle phases are driven primarily by the 0–2 and 2–4 month groups; older age groups do not differ consistently from one another. Selected pairwise comparisons shown ($*p < 0.05$, $***p < 0.001$).

Behavior during probabilistic execution of the two-armed bandit task across the lifespan

Having established that mice across the lifespan acquired the task structure and remained engaged, we next asked whether aging affects flexible updating during the probabilistic task. Under 80/20 contingencies, reward is uncertain at both ports throughout each block, such that identifying the high-reward port and detecting reversals requires integration of outcome history across trials. Performance thus serves as a sensitive measure of ongoing belief updating rather than simple stimulus-response learning.

Older mice accumulated more unrewarded outcomes across sessions, without a corresponding increase in rewarded trials (Fig. 2a). Consistent with this, high-port selection probability was reduced in the oldest age group (Fig. 2c). Trial duration remained stable across older age groups despite elevated error rates (Fig. 2b), indicating that, as during acquisition, the performance deficit reflects erroneous action selection rather than impaired execution.

To explore whether age-related performance deficits were driven by blunted detection of contingency change or diminished post-reversal adaptation, behavior was aligned to block transitions (Fig. 2d). All age groups showed progressive recovery of high-port selection following reversal, indicating that the ability to eventually track contingency changes is preserved across the lifespan. However, pre-reversal baseline high-port selection showed a numerical decline across age groups, from approximately 0.74 in younger mice to 0.66 in the oldest group, and trials required to recover to baseline showed a corresponding numerical increase from approximately 7.6 to 12.7 in the oldest group. Post-reversal switching probability similarly showed a numerical reduction in 18-24 month mice. These trends were consistent in direction but did not reach significance, likely owing to sample size (larger cohorts may be needed to resolve statistically).

To further characterize outcome-dependent updating, we quantified lose-switch and win-stay probabilities as trial-level measures of how outcomes influenced subsequent choices (Figs. 2f, 2g). Lose-switch probability showed a numerical decline with age, from approximately 0.25 in younger groups to 0.20 in the oldest group, while win-stay probability remained stable across age groups. This directional dissociation suggests a selective impairment in updating following informative outcomes rather than a global reduction in choice consistency, though neither difference reached significance.

Together, these findings indicate that aging selectively impairs reinforcement-guided updating under probabilistic conditions, with older mice showing reduced choice optimality and directional evidence of diminished outcome-driven updating across multiple behavioral measures.

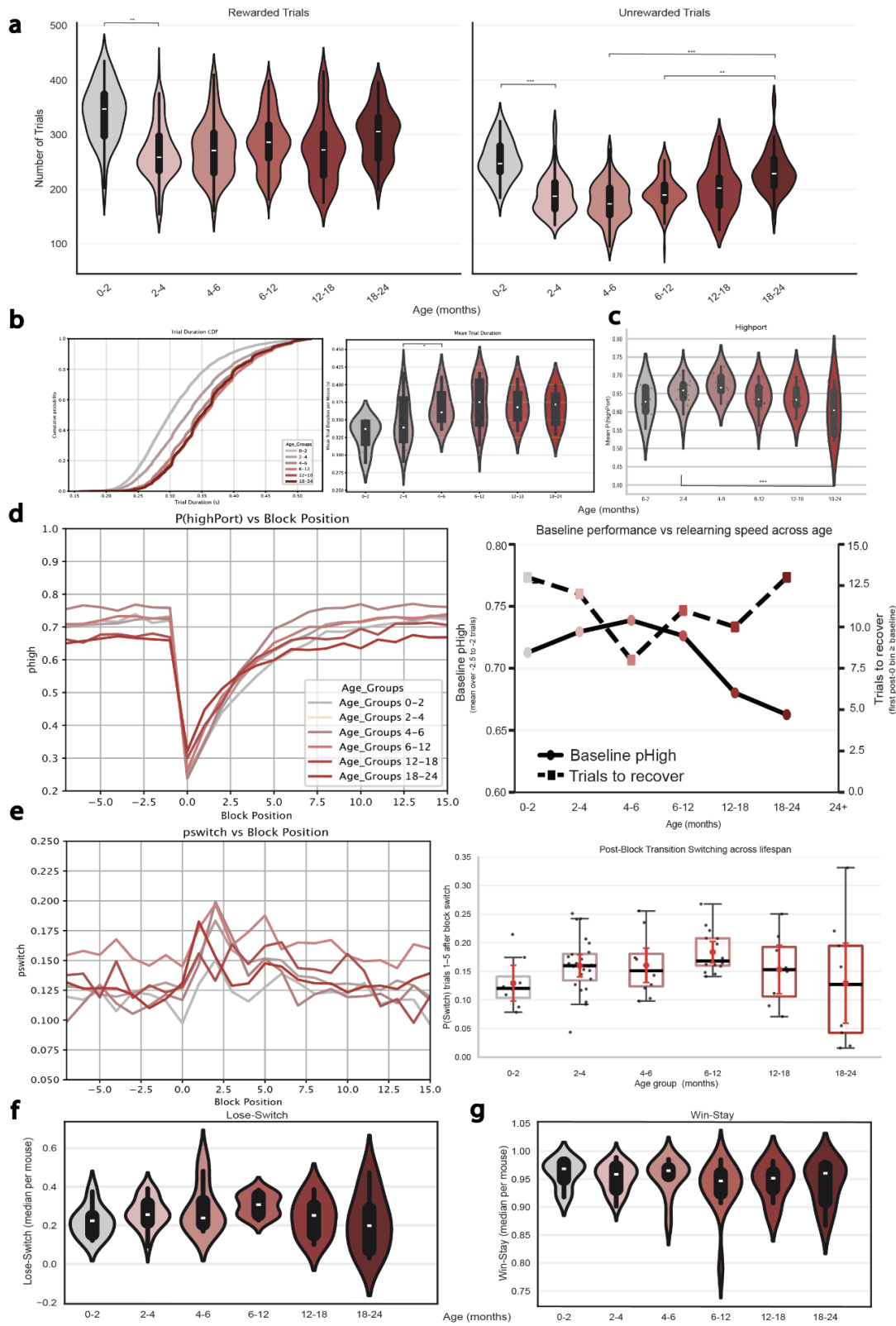


Figure 2. Aging impairs reinforcement-dependent updating during probabilistic decision-making.

n = 8, 24, 11, 15, 8, 9 mice for age groups 0–2, 2–4, 4–6, 6–12, 12–18, and 18–24 months, respectively. **p* < 0.05, ***p* < 0.01, ****p* < 0.001, Tukey HSD post-hoc following one-way ANOVA unless otherwise stated.

a, Rewarded (left) and unrewarded (right) trial counts per session across age groups, plotted as violin plots with embedded box plots (median, IQR, whiskers = 1.5 × IQR). Rewarded trials: $F(5,69) = 3.76$, $p = 0.005$; significant difference driven by the 0–2 month group. Unrewarded trials: $F(5,69) = 12.01$, $p < 0.001$. Selected pairwise comparisons shown (***p* < 0.01, ****p* < 0.001).

b, Trial duration across age groups. Left, cumulative distribution functions per age group. **Right**, mean trial duration per mouse plotted as in a. One-way ANOVA: $F(5,69) = 3.99$, $p = 0.003$; differences driven by the 0–2 month group being slower than all others, with no consistent differences among groups aged 4 months and older. Selected pairwise comparisons shown (***p* < 0.01, ****p* < 0.001).

c, High-port selection probability per mouse across age groups, plotted as in a. One-way ANOVA: $F(6,69) = 3.99$, $p = 0.003$. Selected pairwise comparisons shown (***p* < 0.01).

d, High-port selection aligned to contingency reversal. Left, mean high-port selection probability per age group aligned to reversal (block position 0). **Right**, pre-reversal baseline high-port selection (solid line) and trials required to recover to baseline (dashed line) per age group, plotted as in a. Baseline high-port selection: $F(5,69) = 2.14$, $p = 0.071$. Post-reversal recovery trials: $F(5,68) = 0.70$, $p = 0.623$. No significant pairwise differences detected.

e, Switch probability aligned to contingency reversal. Left, mean switch probability per age group aligned to reversal. **Right**, switch probability in trials 1–5 following reversal per mouse, plotted as in a. One-way ANOVA: $F(5,69) = 1.20$, $p = 0.319$. No significant differences detected.

f, Lose-switch probability per mouse across age groups, plotted as in a. One-way ANOVA: $F(5,69) = 1.58$, $p = 0.177$. No significant differences detected.

g, Win-stay probability per mouse across age groups, plotted as in a. One-way ANOVA: $F(5,69) = 0.86$, $p = 0.512$. No significant differences detected.

Interpretable modelling of probabilistic updating with a recursively formulated logistic regression (RFLR) model

To identify specific computational processes underlying the observed age-dependent changes in behavior, we fitted trial-by-trial choices using a compact reinforcement learning framework previously developed for this task and shown to accurately capture both port selection and switching dynamics (RFLR; Beron et al., 2022; Fig. 3a). In this model, choice probability is generated by applying a stochastic policy to a latent decision variable updated recursively on each trial. Following each outcome, this state decays toward baseline and is adjusted according to the most recent choice and its result, such that rewarded actions reinforce the selected port while unrewarded outcomes shift evidence toward the alternative. Together these dynamics

capture both reinforcement-dependent updating and the stochastic exploration observed in behavior.

The model is defined by three parameters: the persistence parameter α captures “stickiness” or the tendency to repeat the previous choice independent of reinforcement; the reinforcement sensitivity parameter β scales the influence of the immediately previous outcome history on subsequent choice; and the integration timescale parameter τ reflects the temporal persistence of accumulated evidence across trials.

The model accurately reproduced key features of behavior, including switching probabilities and block-aligned choice dynamics (Fig. 3b). Model fit quality remained consistent across age groups, with log likelihood per trial ranging from approximately -0.30 to -0.36, indicating that the model provided a comparably good description of behavior across the lifespan (Fig. 3f).

Inspection of individual model parameters revealed a dissociation across age groups. Parameter α remained stable across age (Fig. 3c), indicating preserved baseline choice “stickiness”. In contrast, β declined significantly with age, with the most pronounced reduction observed in mice aged 18-24 months relative to the 4-6 month group (Fig. 3d). τ increased significantly with age, with 18-24 month mice showing substantially longer integration timescales than younger adults (Fig. 3e). This prolonged persistence of prior evidence likely slows the incorporation of new reinforcement signals, compounding the effect of reduced β on behavioral updating.

Together, these parameter changes identify a distinct computational signature of aging defined by reduced reinforcement sensitivity and increased persistence of prior evidence. This combination could reasonably produce slower behavioral updating and increased stability of prior choices,

accounting for the observed reduction in lose-switch probability, diminished high-port selection, and trend toward delayed post-reversal adaptation.

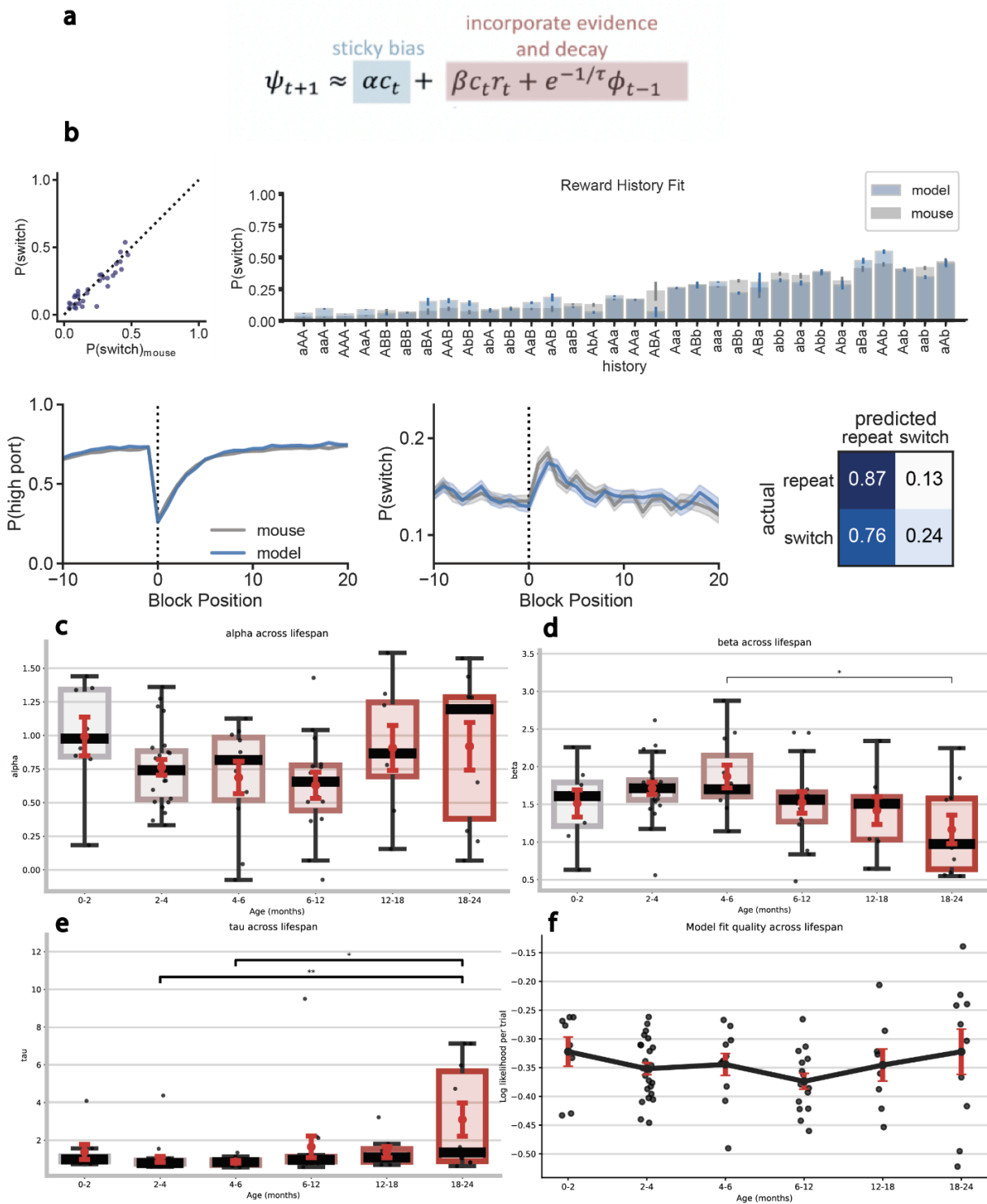


Figure 3. Model reveals age-dependent reduction in reinforcement sensitivity and prolonged persistence of prior evidence.

n = 8, 24, 11, 15, 8, 10 mice for age groups 0–2, 2–4, 4–6, 6–12, 12–18, and 18–24 months, respectively. **p* < 0.05, ***p* < 0.01, Tukey HSD post-hoc following one-way ANOVA unless otherwise stated.

a, RFLR model schematic. The decision variable ψ is updated recursively on each trial as the sum of a sticky bias term (blue) and a reinforcement and decay term (red), where α governs choice persistence, β scales reinforcement sensitivity, and τ controls the timescale of evidence integration.

b, Model validation. **Top left**, predicted versus observed switch probability per mouse across all data. **Top right**, switch probability conditioned on the three most recent action-outcome histories (model, blue; mouse, grey), plotted as mean $\pm$ SEM. **Bottom**, mean high-port selection probability (left) and switch probability (right) aligned to contingency reversal for model and mouse. **Right**, confusion matrix showing predicted versus actual repeat and switch choices across all data.

c, Persistence parameter (α) per mouse across age groups, plotted as box plots (median, IQR, whiskers = $1.5 \times$ IQR) with individual animals overlaid. One-way ANOVA: $F(5,70) = 1.43$, $p = 0.224$. No significant differences detected.

d, Reinforcement sensitivity parameter (β) per mouse across age groups, plotted as in c. One-way ANOVA: $F(5,70) = 2.68$, $p = 0.029$. Selected pairwise comparisons shown (* $p < 0.05$).

e, Evidence integration timescale (τ) per mouse across age groups, plotted as in c. One-way ANOVA: $F(5,70) = 3.10$, $p = 0.014$. Selected pairwise comparisons shown (* $p < 0.05$, ** $p < 0.01$).

f, Model fit quality (log likelihood per trial) per mouse across age groups, plotted as in c, with mean $\pm$ SEM overlaid (red) and polynomial fit (blue). One-way ANOVA: $F(5,70) = 0.87$, $p = 0.504$; Spearman $\rho = -0.037$, $p = 0.753$. No significant differences detected.

Dopamine transients encode reward history in deterministic and probabilistic versions of the two-armed bandit task.

Having identified an age-dependent reduction in reinforcement sensitivity (β) via computational modeling, we next sought to examine a candidate neural substrate through which this change might be expressed. Nucleus accumbens (NAc) dopamine is a well-established encoder of reward prediction errors and a key substrate for reinforcement-guided learning. If NAc dopamine encodes task-relevant information, age-related variation in this signal could plausibly contribute to the behavioral phenotype observed in older animals.

To examine NAc dopamine dynamics during task performance and determine whether they vary across age, we recorded dopamine using the genetically encoded sensor GRABDA3.0 and fiber photometry in head-fixed mice performing the task at P60 (2 months) and P120 (4 months) (Fig. 4a, b). These ages were selected to capture the window in which reward accumulation and trial duration first diverge significantly in the freely moving task (Fig. 2a, b).

Dopamine transients were tightly time-locked to reward consumption and were systematically scaled by recent reward history (Fig. 4c). Transients did not respond uniformly to each reward, instead amplitude decreased progressively across consecutive wins in both age groups, suggesting an encoding of the position of reward within a recent outcome sequence rather than its occurrence alone. P120 mice showed larger absolute responses at each sequence position than P60 mice, but directional attenuation across wins was notably present in both groups.

History-dependent encoding extended beyond reward consumption. Dopamine responses at cue onset increased across consecutive win sequences before plateauing (Fig. 4d). Following losses, signals following consumption were suppressed below baseline in proportion to recent outcome history, with P60 mice showing deeper initial suppression than P120 mice (Fig. 4e). Dopamine thus tracks both positive and negative outcome histories in a graded manner, with amplitude at key task epochs reflecting the outcome context in which each event occurred.

Outcome and age-related response magnitude differences were also demonstrated under the 80/20 contingency. Peak responses at the first reward in a sequence were larger in P120 than P60 mice, and the attenuation across subsequent rewards was more pronounced in P60 animals (Fig. 4f).

The graded encoding of outcome history in dopamine amplitude is consistent with a role for this signal in informing updating, motivating examination of its dynamics across the lifespan.

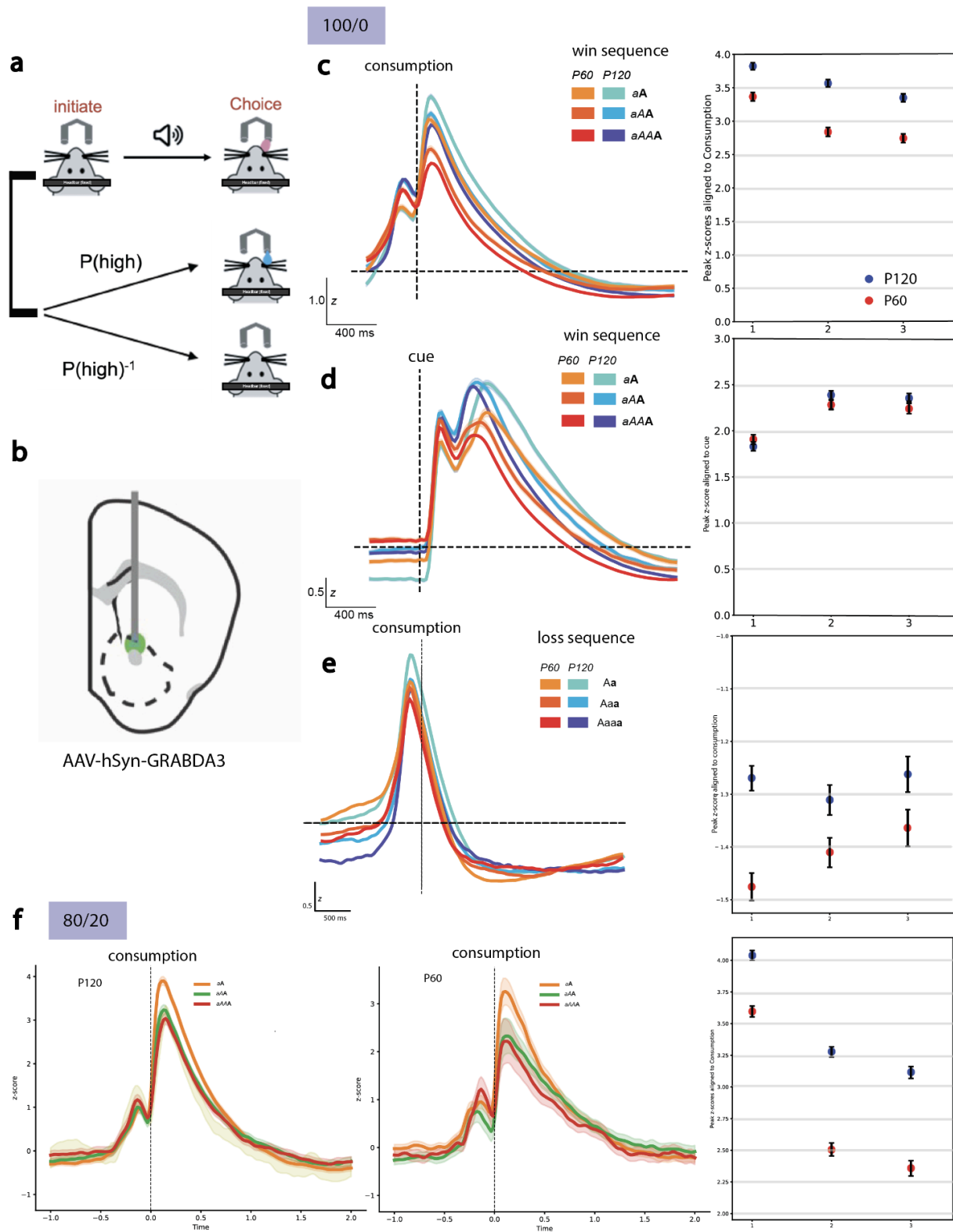


Figure 4. Dopamine transients encode reward history in deterministic and probabilistic versions of the two-armed bandit task.

n = 12, 10 mice for P60 and P120 respectively.

a, Head-fixed task schematic. Head-fixed mice initiated trials by licking the center spout, followed by a tone cue prompting choice of one of two side spouts. Reward was delivered with probability $P(\text{high})$ at the chosen port or withheld with probability $1-P(\text{high})$.

b, Viral delivery strategy. AAV-hSyn-GRABDA3 was injected into the nucleus accumbens for fluorescence-based dopamine imaging.

c, Reward-evoked dopamine responses across win sequences (100/0 contingency). **Left**, z-scored photometry signals aligned to reward consumption for increasing win sequence lengths (aA, aAA, aAAA) in P60 (warm colours) and P120 (cool colours) mice, plotted as mean $\pm$ SEM. **Right**, peak z-scores at each sequence position for P60 (red) and P120 (blue), plotted as mean $\pm$ SEM. P120 peak responses declined from 3.82 ± 0.05 at the first reward in a sequence to 3.34 ± 0.06 at the third; P60 responses declined from 3.36 ± 0.06 to 2.74 ± 0.07 .

e, Consumption-aligned dopamine responses across loss sequences (100/0 contingency). **Left**, z-scored photometry signals aligned to reward consumption for increasing loss sequence lengths (Aa, Aaa, Aaaa) in P60 and P120 mice, plotted as mean $\pm$ SEM. **Right**, peak z-scores at each sequence position plotted as c. Peak suppression following the first loss in a sequence was -1.47 ± 0.03 in P60 mice and -1.27 ± 0.02 in P120 mice.

f, Reward-evoked dopamine responses across win sequences (80/20 contingency). Z-scored photometry signals aligned to reward consumption for P60 (left) and P120 (right) mice across win sequence lengths (aA, aAA, aAAA), plotted as mean $\pm$ SEM. **Right**, peak z-scores at each sequence position plotted as c. Peak responses at the first reward in a sequence were 4.05 ± 0.05 in P120 mice and 3.60 ± 0.05 in P60 mice.

Aging reduces tonic dopamine baseline in the nucleus accumbens

To understand whether and how dopaminergic signaling changes across the lifespan, we used fluorescence lifetime photometry in freely moving mice expressing the genetically encoded dopamine sensor chrimsonDlight (Fig. 5a). Unlike intensity-based photometry, fluorescence lifetime provides a concentration-dependent readout that is largely independent of motion artifacts, photobleaching, and variability in expression or optical coupling, enabling stable comparison of absolute dopamine levels across animals and timepoints.

To verify that the fluorescence lifetime signal reflects biologically meaningful activity, we pharmacologically perturbed dopaminergic signaling via systemic administration of the D1 receptor antagonist SCH23390. Antagonist administration produced a rapid and sustained reduction in fluorescence lifetime to a stable lower bound of approximately 1.7 ns, in line with previously described sensor floor conditions (Fig. 5b; Lodder et. al 2025). Reduction was accompanied by catatonic behavior, confirming effective pharmacological suppression.

Across recordings, fluorescence lifetime signals exhibited a stable tonic baseline punctuated by discrete phasic transients, which were clearly visible as upward deflections in representative traces and differed markedly in their absolute range between young and aged animals (Fig. 5c).

Quantification of baseline lifetime using the modal signal value revealed a significant age-dependent decline, with values remaining stable across early and mid-adulthood before falling substantially in mice aged 24 months and older (Fig. 5d). Peak lifetime values showed a parallel reduction in the oldest group (Fig. 5e).

Phasic amplitudes, quantified as the difference between peak and baseline lifetime, remained stable across all age groups (Fig. 5f), indicating that while the absolute level of dopamine decreases with age, the capacity for phasic release is preserved.

To exclude the possibility that lifetime measurements were confounded by tissue autofluorescence or optical artifacts, we quantified fluorescence intensity as a function of excitation power across animals. Fluorescence intensity scaled linearly with excitation power across the tested range (Fig. 5g), consistent with fluorophore-dependent emission rather than nonlinear autofluorescence or hemodynamic contributions.

Together, these findings demonstrate that aging is associated with a selective reduction in tonic dopamine baseline in the NAc, such that phasic transients occur from a lower absolute starting point while retaining their relative amplitude. Given that NAc dopamine encodes reward history in a graded manner (Fig. 4), a reduction in the absolute range of this signal provides a plausible mechanism for the diminished reinforcement sensitivity observed in older animals.

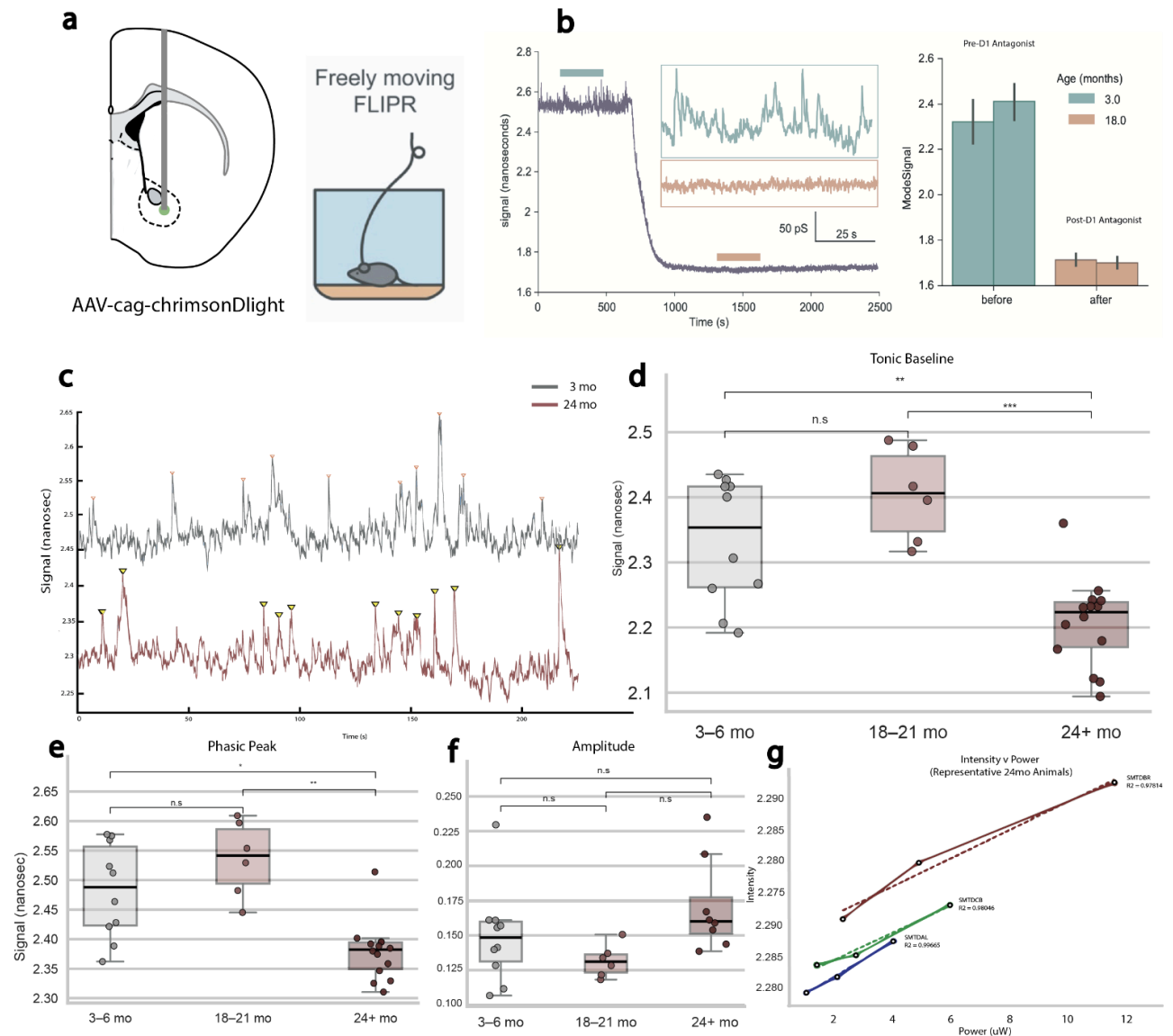


Figure 5. Aging produces a downward shift in tonic dopamine baseline while preserving relative phasic signaling.

$n = 10, 6,$ and 14 recordings from $4, 3,$ and 5 mice at $3-6, 18-21,$ and $24+$ months respectively. Statistical unit is recording. Several animals contributed recordings at multiple timepoints. $*p < 0.05, **p < 0.01, ***p < 0.001$, Tukey HSD post-hoc following one-way ANOVA unless otherwise stated.

a, Surgical and recording schematic. AAV-cag-chrimsonDlight was injected into the nucleus accumbens and fluorescence lifetime was recorded in freely moving mice using a FLIPR fiber photometry system.

b, Pharmacological validation. Left, representative fluorescence lifetime trace during systemic administration of the D1 receptor antagonist SCH23390 (teal bar, 3 months; orange bar, 18 months). Inset, expanded traces showing signal dynamics in each age group before and after antagonist administration. Right, modal lifetime signal before and after antagonist administration in 3 month (teal) and 18 month (orange) mice, plotted as mean $\pm$ SEM.

c, Representative lifetime traces across age. Overlay of fluorescence lifetime recordings from a 3 month (grey) and 24 month (red) mouse during free behaviour. Triangles denote detected phasic transient events.

d, Baseline lifetime across age groups. Modal baseline fluorescence lifetime per recording plotted as box plots (median, IQR, whiskers = $1.5 \times \text{IQR}$) with individual recordings overlaid. One-way ANOVA: $F(2,21) = 9.80$, $p < 0.001$. Baseline lifetime was significantly reduced in 24+ month recordings relative to 3–6 months ($*p = 0.014$, Tukey HSD); 18–21 and 3–6 month groups did not differ significantly ($p = 0.269$). Selected pairwise comparisons shown ($*p < 0.05$, $***p < 0.001$).

e, Peak lifetime across age groups. Peak fluorescence lifetime per recording plotted as in d. One-way ANOVA: $F(2,21) = 9.47$, $p = 0.001$. Peak lifetime was significantly reduced in 24+ month recordings relative to both 18–21 months ($**p = 0.001$, Tukey HSD) and 3–6 months ($*p = 0.013$, Tukey HSD); 18–21 and 3–6 month groups did not differ significantly ($p = 0.329$). Selected pairwise comparisons shown ($*p < 0.05$, $**p < 0.01$).

f, Phasic amplitude across age groups. Peak minus baseline fluorescence lifetime per recording plotted as in d. One-way ANOVA: $F(2,21) = 2.97$, $p = 0.073$. No significant differences detected.

g, Autofluorescence control. Fluorescence intensity plotted as a function of excitation power for representative 24 month animals, with linear regression fits and R^2 values shown ($R^2 = 0.978, 0.980, 0.997$).

Tonic dopamine baseline does not covary with motivational state or effort expenditure

A key alternative interpretation of the age-related reduction in tonic dopamine baseline is that it reflects a generalised change in motivational state. If older animals are simply less driven to seek reward, this alone could account for slower acquisition and reduced behavioral sensitivity to outcomes without implicating the computation itself.

To assess motivation separately, mice were trained on a progressive ratio schedule in which the number of responses required per pellet increased incrementally across the session (Fig. 6a; See Materials and Methods).

Breakpoint on the PR task declined with increasing percentage of initial body weight, confirming that the task was sensitive to motivational state, with less food-restricted animals ceasing to respond at lower ratio requirements (Fig. 6b).

Breakpoint, however, did not differ across age groups, indicating that motivational effort is broadly stable across the lifespan.

To determine whether individual differences in tonic dopamine baseline predicted effort expenditure, we examined the relationship between dopamine lifetime measures and breakpoint across animals. Neither baseline nor peak dopamine lifetime was significantly associated (Fig. 6d). To further test this, we examined hungry versus sated sessions. Baseline lifetime did not differ significantly between hunger states in any age group (Fig. 6e), indicating that tonic dopamine tone is not acutely modulated by hunger state.

Together, these findings demonstrate that the age-related reduction in tonic dopamine baseline is dissociable from both motivational effort and hunger state, suggesting that this decline more plausibly impacts reinforcement signaling than it reflects a global reduction in motivational drive.

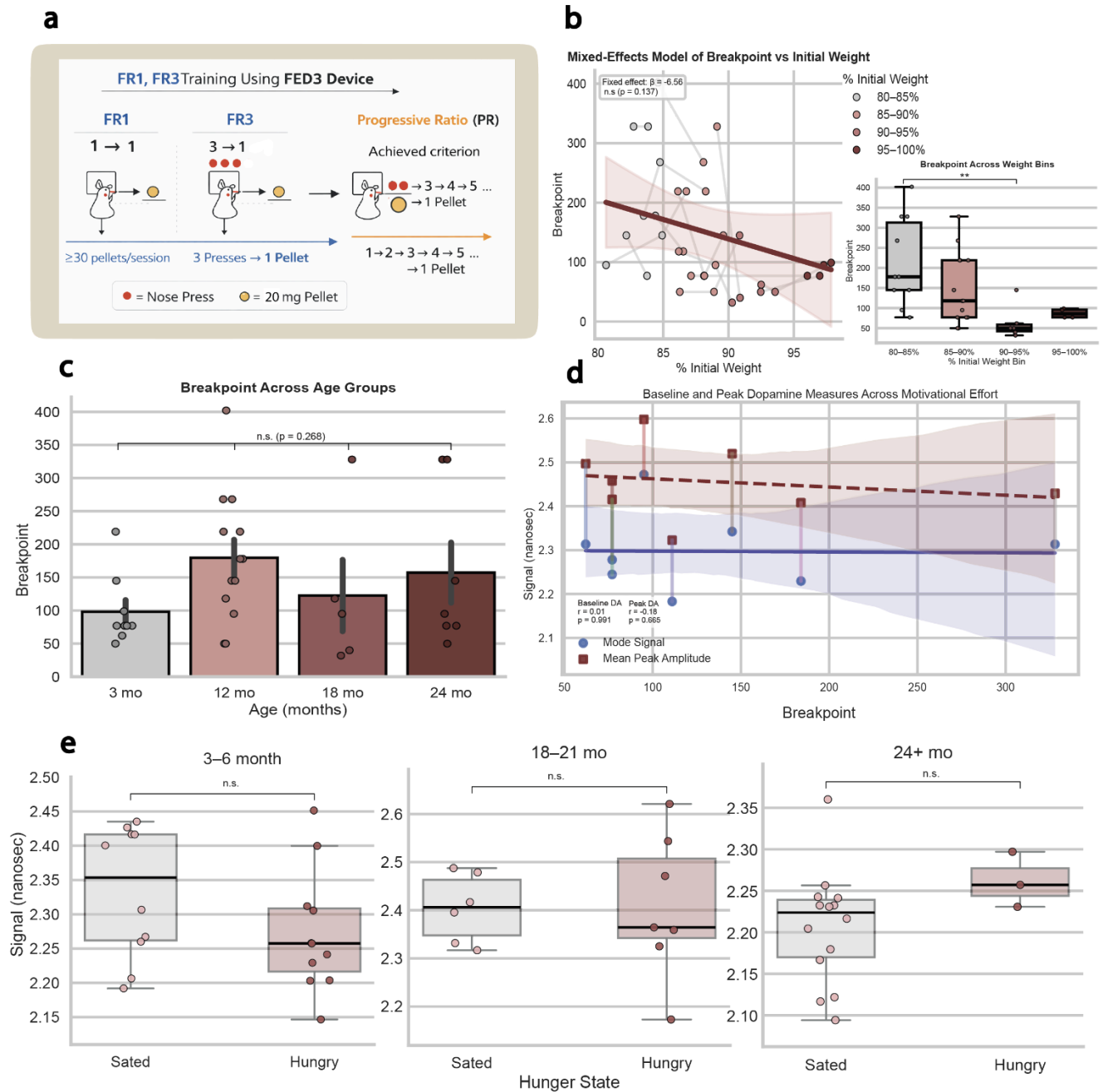


Figure 6. Tonic dopamine baseline does not predict motivation or hunger state across aging.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, Tukey HSD post-hoc following one-way ANOVA unless otherwise stated.

a, Progressive ratio task schematic. Mice were trained on fixed ratio schedules (FR1, FR3) using a FED3 device before progressing to a progressive ratio schedule in which the number of nose presses required per pellet increased incrementally across the session. Breakpoint was defined as the highest ratio completed before cessation of responding.

b, Breakpoint as a function of body weight. Left, PR breakpoint plotted against percentage of initial body weight per recording, with mixed effects model fit and individual animal trajectories shown. Right, breakpoint across percentage initial

weight bins plotted as box plots (median, IQR, whiskers = $1.5 \times \text{IQR}$) with individual recordings overlaid. One-way ANOVA: $F(3,30) = 4.65$, $p = 0.009$. Selected pairwise comparisons shown ($**p < 0.01$). $n = 34$ recordings from 13 mice.

c, Breakpoint across age groups. PR breakpoint per recording plotted as box plots with individual recordings overlaid across 3, 12, 18, and 24 month age groups. One-way ANOVA: $F(3,30) = 1.38$, $p = 0.268$. No significant differences detected. $n = 9, 13, 5, 7$ recordings at 3, 12, 18, and 24 months respectively.

d, Dopamine lifetime measures as a function of PR breakpoint. Baseline dopamine lifetime (mode signal, blue) and mean peak amplitude (dark red) plotted against PR breakpoint per animal, with linear regression fits and 95% confidence intervals. Neither baseline (Pearson $r = 0.005$, $p = 0.991$) nor peak dopamine (Pearson $r = -0.183$, $p = 0.665$) predicted breakpoint. $n = 8$ animals.

e, Baseline dopamine lifetime in sated and hungry states across age groups. Modal baseline lifetime per session plotted as box plots with individual sessions overlaid for 3–6 month (left), 18–21 month (middle), and 24+ month (right) animals in sated and hungry conditions. No significant difference between hunger states was detected in any age group (3–6 months: Mann–Whitney $U = 39.0$, $p = 0.176$; 18–21 months: $U = 45.0$, $p = 0.364$; 24+ months: $U = 29.0$, $p = 0.136$). $n = 12$ hungry and 10 sated sessions at 3–6 months; $n = 7$ hungry and 10 sated sessions at 18–21 months; $n = 3$ hungry and 12 sated sessions at 24+ months.

Aging drives coordinated transcriptional remodeling across striatal cell types.

The drivers of age-related reduction in tonic dopamine remain unresolved. One possibility is that it reflects broader remodeling of the striatal environment within which dopamine signals operate. To characterize this landscape, we performed a hypothesis-free re-analysis of single-nucleus RNA sequencing data from Linker et al. (2025), examining age-dependent gene expression changes across all major striatal cell populations.

Unsupervised clustering identified ten transcriptionally distinct populations in both age groups (Fig. 7a). Two patterns in the compositional and burden analyses motivated deeper examination of specific cell types. Astrocytes showed the largest proportional decline with age yet carried the lowest differential expression burden, indicating state compression of the surviving population rather than transcriptional activation. Microglia demonstrated the opposite, featuring a modest proportional increase accompanied by the greatest transcriptional disruption across cell types, exceeding 2,000 differentially expressed genes (Fig. 7b, c). Spiny projection neurons, as the principal output neurons of the striatum and the direct cellular substrate through which dopamine

signals are translated into behavioral updating, were examined for age-dependent transcriptional changes. Striatal interneurons were additionally examined given their proposed role in regulating striatal circuit gain and dopamine release probability.

Microglia exhibited coordinated loss of homeostatic markers *P2ry12*, *Tmem119*, *Sall1*, *Hexb*, and *Cx3cr1* with concurrent gain of DAM markers *Trem2*, *Apoe*, *Lpl*, *Cst7*, and *Axl* (Fig. 7d).

Among surviving astrocytes, *Vim* and *S100b* were significantly elevated while *Slc1a2*, encoding the primary astrocytic glutamate transporter GLT-1, was significantly reduced (Fig. 7e), suggesting that aged astrocytes are both fewer and functionally compromised in their capacity to buffer extracellular glutamate.

Striatal interneurons and spiny projection neurons displayed convergent upregulation of mitochondrially encoded transcripts including *Lars2*, *Gm42418*, *mt-Co2*, *mt-Nd3*, and *mt-Atp8*, alongside downregulation of genes involved in epigenetic regulation, proteostasis, and synaptic signaling (Fig. 7f, g).

Together, these findings reveal a broadly remodeled striatal environment in aged mice, characterized by reactive microglial activation, compromised astrocytic glutamate buffering, and convergent bioenergetic stress across neuronal populations. Each of these changes would be expected to degrade the fidelity of dopamine signaling and impair synaptic machinery, providing a molecular framework that complements our identified physiological and computational signatures of aging.

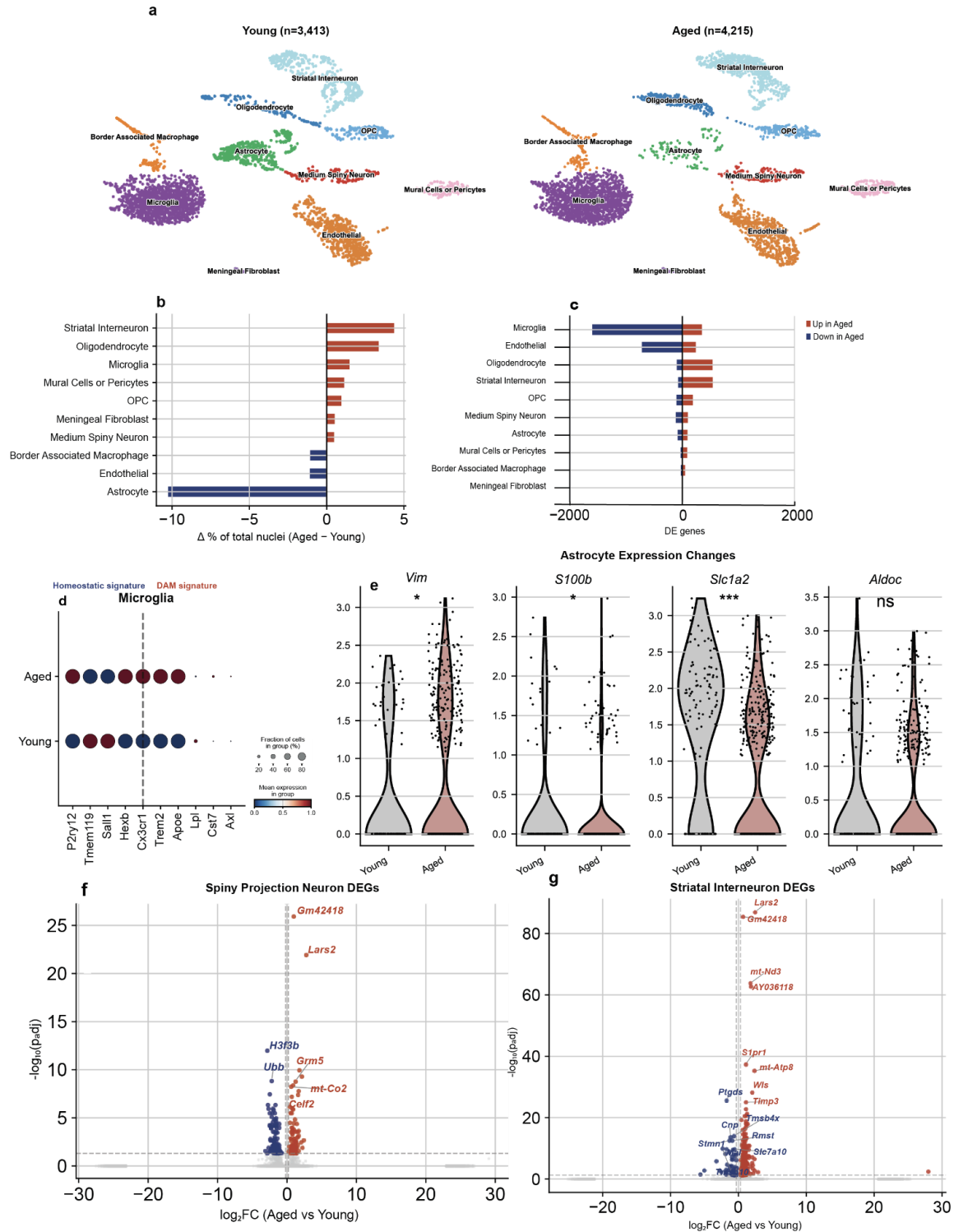


Figure 7. Aging drives coordinated transcriptional remodeling across striatal cell types.

Data derived from reanalysis of deposited single-nucleus RNA sequencing data from Linker et al. (2025), Nature Communications 16, 8496. Young: 2 months; aged: 18 months. $n = 4$ mice per group. * $p < 0.05$, *** $p < 0.001$, Mann-Whitney U test unless otherwise stated.

a, Single-nucleus transcriptomic atlas of young and aged striatum. UMAPs of nuclei from young (left, n = 3,413) and aged (right, n = 4,215) mouse striatum colored by cell type. Ten transcriptionally distinct clusters were identified across both age groups.

b, Cell type proportional composition shifts with age. Change in percentage of total nuclei per cell type ($\Delta\%$ = aged minus young). Astrocytes show the largest proportional decline; striatal interneurons show the largest proportional increase.

c, Differential expression burden by cell type. Number of upregulated (red) and downregulated (blue) DEGs per cell type ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 0.2$). Microglia carry the greatest transcriptional burden; astrocytes the least.

d, Microglia homeostatic-to-DAM transition. Dot plot of homeostatic (P2ry12, Tmem119, Sall1, Hexb, Cx3cr1) and DAM (Trem2, Apoe, Lpl, Cst7, Axl) marker expression in young and aged microglia. Dot size reflects fraction of cells expressing the gene; color reflects mean expression level.

e, Astrocyte marker gene expression across age. Violin plots of Vim, S100b, Slc1a2, and Aldoc in young (grey) and aged (salmon) astrocytes (young n = 436, aged n = 105). Vim and S100b are significantly elevated in aged astrocytes ($p = 0.019$ and $p = 0.024$ respectively); Slc1a2 is significantly reduced ($p < 0.001$); Aldoc is unchanged. Selected comparisons shown (* $p < 0.05$, *** $p < 0.001$, Mann-Whitney U test).

f, Spiny Projection neuron differential expression. Volcano plot of SPN DEGs (aged vs young). Top upregulated transcripts include Gm42418, Lars2, and mt-Co2; top downregulated include H3f3b, Ubb, Grm5, and Celf2.

g, Striatal interneuron differential expression. Volcano plot of interneuron DEGs (aged vs young, 539 upregulated, 74 downregulated). Top upregulated transcripts include Lars2, Gm42418, mt-Nd3, and mt-Atp8.

IV. Discussion

In this study, we show that aging is associated with reduced reinforcement sensitivity during adaptive choice and a concurrent downward shift in tonic dopamine baseline in the NAc, implicating tonic dopaminergic tone as a mechanism through which aging alters behavioral updating.

Selective vulnerability of tonic dopamine signaling in aging

A central observation is that tonic dopamine baseline declines with age while phasic signaling dynamics remain preserved. The selective nature of this dissociation suggests that decline is more likely a change specific to the mechanisms that set or maintain extracellular dopamine than global dopaminergic deterioration.

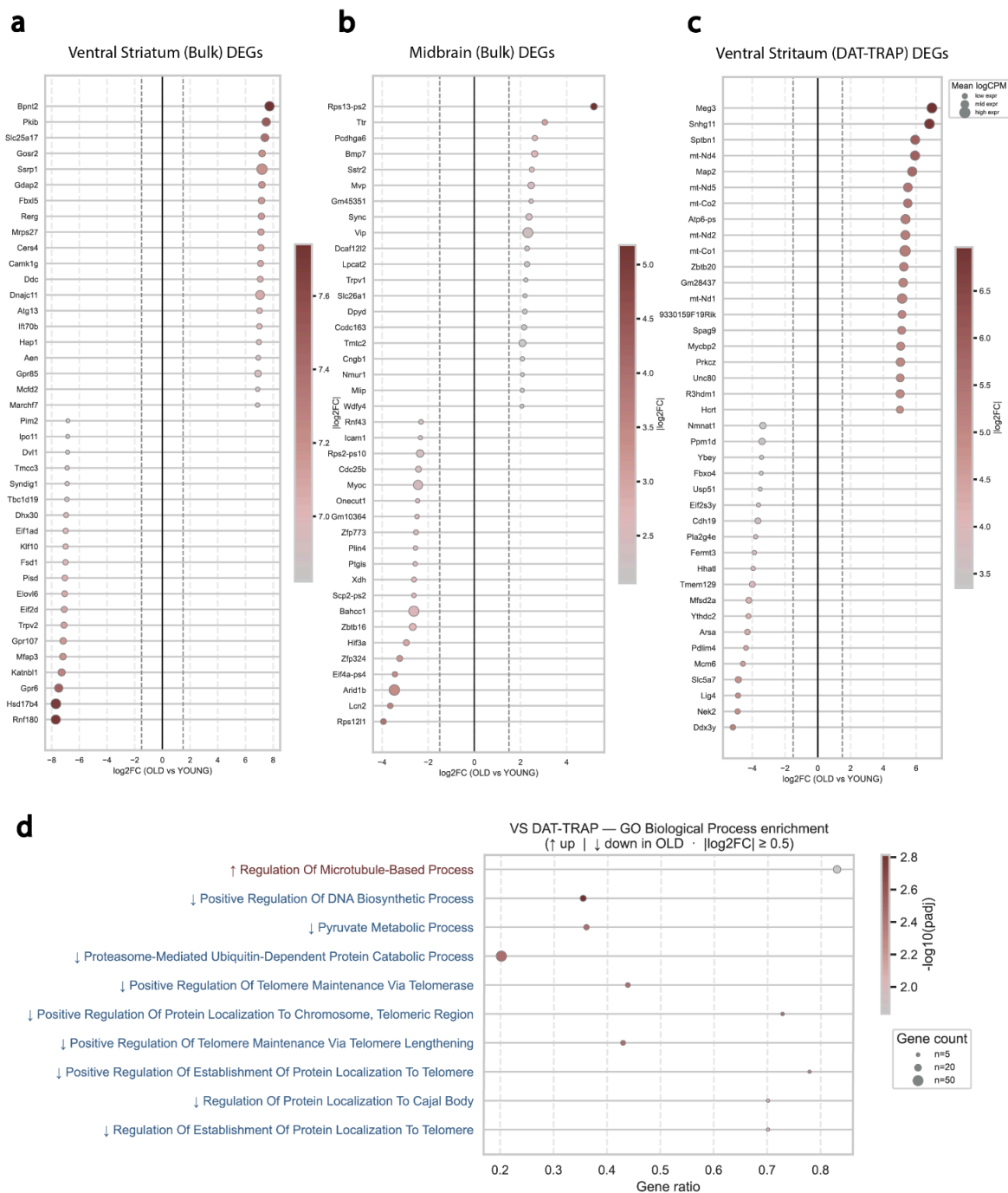
Several mechanisms could contribute. Aging-related cellular changes may alter intrinsic excitability of dopaminergic neurons, for example through changes in ion channel expression or membrane properties that reduce baseline firing rates without impairing burst firing. Cholinergic interneurons are known to exert powerful control over dopamine release probability and could modulate basal dopamine independently of phasic dynamics. Changes in dopamine clearance mechanisms, including altered expression or function of DAT, could further reduce extracellular dopamine levels by accelerating reuptake. Global neuromodulatory or hormonal changes across aging may complementarily alter the excitability landscape of dopaminergic neurons or their afferent inputs. Another intriguing possibility is that neurons sustaining tonic firing may be disproportionately vulnerable to age-related damage owing to their higher resting energetic demands, though this interpretation depends on whether functionally distinct tonic and phasic dopamine neuron populations exist, a question that remains unresolved (Liu et al. 2021).

Exploratory molecular evidence at dopaminergic terminals

Distinguishing between these possibilities requires resolution at the level of the dopaminergic terminal itself. To begin to address this, we performed an exploratory re-analysis of translational data from Kilfeather et al. (2024), which employed a DAT-TRAP strategy to affinity-purify actively translating ribosomes selectively from dopaminergic cell bodies and axonal terminals in the midbrain and ventral striatum respectively. Given the limited replicate numbers in this dataset, findings should only be interpreted as preliminary and hypothesis-generating.

Mitochondrially encoded respiratory chain subunits including mt-Nd1, mt-Nd2, mt-Nd4, mt-Nd5, mt-Co1, and mt-Co2 were among the most consistently upregulated transcripts in aged dopaminergic terminals, suggesting bioenergetic stress within the terminal compartment

consistent with the high energetic demands imposed by sustained vesicle cycling and dopamine reuptake. Among downregulated transcripts was *Slc5a7*, encoding the high-affinity choline transporter, supporting the possibility that cholinergic regulation of dopamine release probability at striatal terminals is altered with age. Interestingly, *Pkib*, encoding the beta isoform of the protein kinase A inhibitor, was upregulated in aged ventral striatum. These observations provide an additional layer to the broader patterns identified across striatal cell types in Figure 7, and together motivate future work with adequate replication to validate these candidates and test their functional relevance through pharmacological, metabolic, or genetic perturbation.



Supplemental Figure 1. Reanalysis of Kilfeather et al. (2024) TRAP sequencing data reveals age-dependent transcriptional remodeling in ventral striatum and midbrain dopaminergic neurons.

Data reanalyzed from Kilfeather et al. (2024), *Cell Reports* 43, 113784. Young: 3 months; aged: 18+ months.

a, Ventral striatum bulk differential gene expression. Top differentially expressed genes in ventral striatum bulk tissue ranked by $|\log_2FC|$ (old vs young), thresholded at $|\log_2FC| \geq 1.5$. Dot size reflects mean logCPM; color intensity reflects $|\log_2FC|$. Note $n = 1$ per condition; results are exploratory.

b, Midbrain bulk differential gene expression. Plotted as a for ventral striatum bulk tissue.

c, DAT-TRAP ventral striatum differential gene expression. Top differentially expressed genes in DAT-TRAP ventral striatum, reflecting translational changes specific to the dopaminergic axonal terminal compartment, thresholded at $|\log_2FC| \geq 1.5$. Mitochondrially encoded transcripts are prominent among genes upregulated in aged animals; *Slc5a7* is among the most consistently downregulated. Note $n = 1$ per condition; results are exploratory.

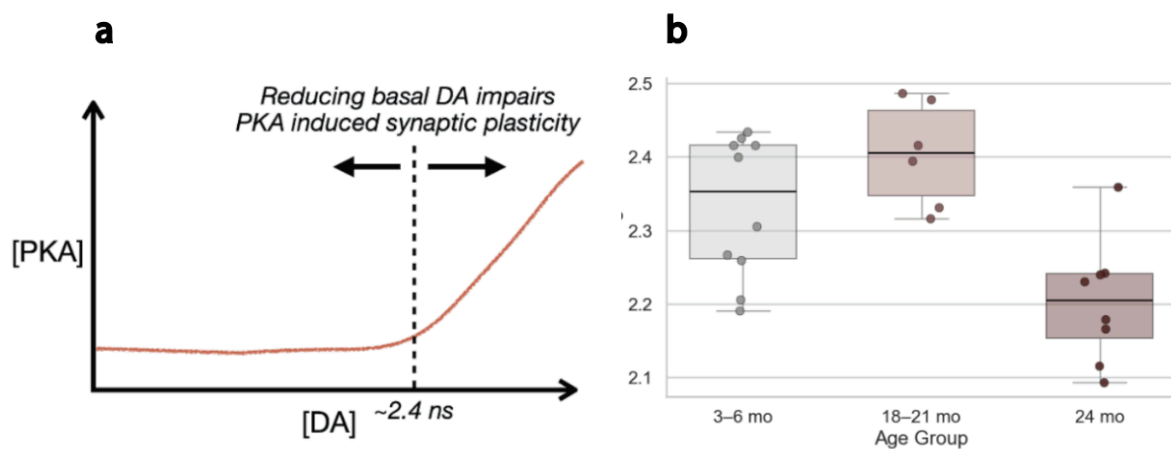
d, Gene Ontology Biological Process enrichment in aged DAT-TRAP ventral striatum terminals. Top enriched GO Biological Process terms from DAT-TRAP differentially expressed genes ($|\log_2FC| \geq 0.5$, adjusted $p < 0.05$). Arrow direction indicates whether the gene set is upregulated ($\uparrow$) or downregulated ($\downarrow$) in aged animals. Dot size reflects gene count; color reflects $-\log_{10}(padj)$. Proteasome-mediated ubiquitin-dependent protein catabolism and pyruvate metabolic process are among the most significantly downregulated pathways; regulation of microtubule-based processes is upregulated.

Functional implications of reduced tonic dopamine baseline

Regardless of the upstream mechanism, a reduction in tonic dopamine baseline carries direct functional consequences for the striatal circuits through which action-outcome is translated into behavioral updating. As we propose in Supplemental Figure 2, reduced tonic baseline shifts the resting state further from the PKA activation threshold, such that equivalent phasic transients produce weaker engagement of D1-SPN and D2-SPN intracellular plasticity pathways. In this view, the reinforcement signal is present but its capacity to drive downstream plasticity is diminished by the condition in which it arrives.

A complementary possibility is that aged SPNs are themselves less able to mount an intracellular response even when dopamine receptor engagement is sufficient. The upregulation of *Pkib* in aged dopaminergic terminals raises the potential that PKA signaling machinery within the terminal may itself be actively suppressed independent of receptor occupancy (Supplementary Figure 1) .

These mechanisms are not mutually exclusive and could compound one another to erode the fidelity with which dopamine signals are transduced into synaptic change. We are actively combining direct dopamine measurements with D1/D2-SPN PKA imaging to determine whether a causative relationship exists and to elucidate potential relative contributions of each.



Supplemental Figure 2: Age-dependent reduction in tonic dopamine baseline shifts signaling further from PKA activation threshold

a, Schematic of striatal dopamine concentration and PKA activation relationship. The PKA activation curve as a function of dopamine concentration is understood to be sigmoidal, with a steep activation region centered around approximately 2.4 ns fluorescence lifetime. A reduction in basal dopamine tone shifts the resting state leftward along this curve, increasing the distance between tonic baseline and the activation threshold and thus requiring larger phasic excursions to drive equivalent PKA-dependent plasticity.

b, Age-dependent decline in tonic dopamine baseline. Modal fluorescence lifetime per animal across age groups (3 to 6, 18 to 21, and 24+ months), plotted as box plots (median, IQR, whiskers = 1.5× IQR) with individual animals overlaid. Baseline lifetime remains stable between 3 to 6 and 18 to 21 months before declining substantially at 24+ months, falling below the approximate PKA activation threshold of 2.4 ns.

A further possibility is that aging alters transcriptional permissiveness in striatal circuits. Even if dopamine transients retain sufficient amplitude to activate receptors and initiate intracellular signaling, the transcriptional machinery required to consolidate reinforcement-dependent updating may become less responsive. Variability in the capacity for stimuli-responsive

transcription has previously been demonstrated with predator odor stress paradigms in mice (Long et al., 2025). The convergent downregulation of genes involved in epigenetic regulation and proteostasis across striatal cell types in Figure 7 lends independent support to this idea. Through this mechanism, reinforcement signals may persist but the ability of striatal circuits to translate them into stable synaptic change may progressively diminish across aging.

Behavioral experience may preserve reinforcement-related circuit function across aging

Whether the trajectory of dopaminergic and cognitive aging is fixed or can be modulated by experience also remains an open question. In an ongoing longitudinal experiment, repeated exposure to the two-armed bandit task preliminarily appears to preserve reinforcement sensitivity across the lifespan, with animals maintaining stable behavioral performance and RFLR model parameters into advanced age compared to age-matched naive controls. Sustained active task experience could stabilize dopaminergic circuit function across multiple levels, from activity-driven plasticity and preserved transcriptional responsiveness to prevention of maladaptive circuit remodeling over longer timescales.

To exclude the possibility of task-specific procedural memory, we are currently testing longitudinally trained animals on a series of learning paradigms through the Harvard Mouse Behavior Core (MBC). If this preservation generalizes across tasks, behavioral engagement may itself represent both a modifiable factor in cognitive aging and a tractable intervention target.

Conclusions and implications

Our findings identify a selective decline in tonic dopamine baseline as a physiological correlate of reduced reinforcement sensitivity across the lifespan. Prior work has documented broad dopaminergic deterioration with age and linked it to impaired reinforcement learning, yet the specific signal component responsible has remained unclear. The present findings begin to resolve this, showing that tonic baseline declines while phasic dynamics are preserved and that this physiological change tracks a corresponding reduction in computational sensitivity to recent outcomes. Integrating behavioral, physiological, and molecular levels of analysis, this work provides a foundation for future studies aimed at establishing causality and identifying the mechanisms that regulate tonic dopamine tone across the lifespan.

These findings also carry direct implications for intervention. Dopaminergic transmission is pharmacologically accessible, and identifying tonic baseline as the affected signal component opens a targeted path toward treatment. If restoring baseline tone in future experiments proves sufficient to rescue reinforcement-dependent updating in aged animals, strategies that elevate tonic dopamine may represent viable approaches to maintaining cognitive flexibility.

Limitations of the study

Limitation 1. Behavioral measurements and fluorescence lifetime recordings were obtained in separate cohorts. As a result, it was not possible to directly test how tonic dopamine baseline relates to reinforcement sensitivity within the same animal. Although both behavioral reinforcement sensitivity and tonic dopamine baseline decline with age, this cohort separation prevents direct assessment of whether individual differences in dopamine baseline predict reinforcement learning parameters or behavioral updating dynamics. Future experiments

combining chronic fluorescence lifetime recordings with 2-ABT behavior in the same implanted animals will be essential for establishing whether tonic dopamine baseline directly tracks reinforcement sensitivity at the individual level. Such experiments would further enable trial-by-trial coupling between dopamine dynamics and behavioral updating to be quantified, providing a more direct test of the proposed mechanistic relationship.

Limitation 2. Present findings establish a physiological correlate of altered reinforcement sensitivity but do not demonstrate causality. Direct manipulation of tonic dopamine levels will be required to determine whether restoring baseline dopamine tone is sufficient to rescue reinforcement-dependent updating in aged animals. A chemogenetic “tickle” approach has previously been successfully demonstrated in the Sabatini lab.

Limitation 3. An initial goal was to directly measure the relationship between tonic dopamine baseline and striatal PKA signaling by co-expressing chromsonDlight and FLIM-AKAR in the nucleus accumbens. Substantial time was invested in optimizing dual viral injection strategies, but reliable co-expression and signal quality sufficient for simultaneous lifetime imaging of both sensors could not be achieved within the timeframe of this thesis. This experiment remains a direct priority.

Limitation 4. The chromsonDlight sensor has not been fully characterized. While lifetime signals reliably reflect changes in dopamine binding to the sensor, additional approaches such as electrochemical recordings or complementary optical sensors will be necessary to map lifetime values to absolute dopamine concentrations *in vivo* and determine what the observed baseline shift represents in physiological terms.

Limitation 5. Dopamine measurements were restricted to the NAcc and it remains unclear whether similar tonic baseline shifts occur across other dopaminergic projection targets, including additional striatal regions (DMS, DLS, VLS), cortical regions (PFC, OFC), or limbic structures (AMY, HIP) which have each been shown to influence 2-ABT gameplay. Because different dopaminergic pathways support distinct aspects of learning and behavior, determining the anatomical specificity of tonic dopamine changes will be important for understanding how aging affects different functionally distinct aspects of the behavior.

V. References

1. Oberman L, Pascual-Leone A (2013) Changes in plasticity across the lifespan: Cause of disease and target for intervention. *Prog Brain Res* 207:91–120.
2. Voglewede RL, Vandemark KM, Davidson AM, DeWitt AR, Heffler MD, Trimmer EH, Mostany R (2019) Reduced sensory-evoked structural plasticity in the aging barrel cortex. *Neurobiol Aging* 81:222–233.
3. Marzola P, Melzer T, Pavesi E, Gil-Mohapel J, Brocardo PS (2023) Exploring the role of neuroplasticity in development, aging, and neurodegeneration. *Brain Sci* 13:1610.
4. Citri A, Malenka RC (2008) Synaptic plasticity: Multiple forms, functions, and mechanisms. *Neuropsychopharmacol* 33:18–41.
5. Pan Y, Monje M (2020) Activity shapes neural circuit form and function: A historical perspective. *J Neurosci* 40:944–954.
6. Bozorgmehr T, Ardiel EL, McEwan AH, Rankin CH (2013) Mechanisms of plasticity in a *Caenorhabditis elegans* mechanosensory circuit. *Front Physiol* 4:88.
7. Sasakura H, Mori I (2013) Behavioral plasticity, learning, and memory in *C. elegans*. *Curr Opin Neurobiol* 23:92–99.
8. Favila N, Capece Marsico J, Pacheco CM, et al. (2025) Heterogeneous plasticity of amygdala interneurons in associative learning and extinction. *Nat Commun* 16:9926.
9. Hille M, Kühn S, Kempermann G, Bonhoeffer T, Lindenberger U (2024) From animal models to human individuality: Integrative approaches to the study of brain plasticity. *Neuron* 112:3522–3541.
10. Herzberg MP, Nielsen AN, Luby J, et al. (2025) Measuring neuroplasticity in human development: The potential to inform the type and timing of mental health interventions. *Neuropsychopharmacol* 50:124–136.

11. Whyte AJ, Kietzman HW, Swanson AM, Butkovich LM, Barbee BR, Bassell GJ, Gross C, Gourley SL (2019) Reward-related expectations trigger dendritic spine plasticity in the mouse ventrolateral orbitofrontal cortex. *J Neurosci* 39:4595–4605.
12. Breton YA, Seeland KD, Redish AD (2015) Aging impairs deliberation and behavioral flexibility in inter-temporal choice. *Front Aging Neurosci* 7:41.
13. Schoenbaum G, Nugent S, Saddoris MP, Gallagher M (2002) Teaching old rats new tricks: Age-related impairments in olfactory reversal learning. *Neurobiol Aging* 23:555–564.
14. Zhang J, Liu D, Fu P, Liu Z-Q, Lai C, Yang C-Q, Chen K, Bao W-D, Hu F, Du H-Y, Yang W, Wang J, Man H-Y, Lu Y, Zhu L-Q (2022) Social isolation reinforces aging-related behavioral inflexibility by promoting neuronal necroptosis in basolateral amygdala. *Mol Psychiatry* 27:4050–4063.
15. Tait DS, Chase EA, Brown VJ (2013) Tacrine improves reversal learning in older rats. *Neuropharmacology* 73:284–289.
16. Norris CM, Korol DL, Foster TC (1996) Increased susceptibility to induction of long-term depression and long-term potentiation reversal during aging. *J Neurosci* 16:5382–5392.
17. Foster TC, Kumar A (2007) Susceptibility to induction of long-term depression is associated with impaired memory in aged Fischer 344 rats. *Neurobiol Learn Mem* 87:522–535.
18. Ducote AL, Voglewede RL, Mostany R (2024) Dendritic spines of layer 5 pyramidal neurons of the aging somatosensory cortex exhibit reduced volumetric remodeling. *J Neurosci* 44:e1378242024.
19. Boric K, Muñoz P, Gallagher M, Kirkwood A (2008) Potential adaptive function for altered long-term potentiation mechanisms in aging hippocampus. *J Neurosci* 28:8034–8039.
20. Wang Y (2008) Differential effect of aging on synaptic plasticity in the ventral and dorsal striatum. *Neurobiol Learn Mem* 89:70–75.
21. Chepkova AN, Schönfeld S, Sergeeva OA (2015) Age-related alterations in the expression of genes and synaptic plasticity associated with nitric oxide signaling in the mouse dorsal striatum. *Neural Plast* 2015:458123.
22. Pauwels L, Chalavi S, Swinnen SP (2018) Aging and brain plasticity. *Aging (Albany NY)* 10:1789–1790.
23. James CE, Müller DM, Müller CAH, Van De Looij Y, Altenmüller E, Kliegel M, Van De Ville D, Marie D (2024) Randomized controlled trials of non-pharmacological interventions for healthy seniors: Effects on cognitive decline, brain plasticity and activities of daily living—A 23-year scoping review. *Heliyon* 10:e26674.
24. Speranza L, di Porzio U, Viggiano D, de Donato A, Volpicelli F (2021) Dopamine: The neuromodulator of long-term synaptic plasticity, reward, and movement control. *Cells* 10:735.
25. Pignatelli M, Bonci A (2015) Role of dopamine neurons in reward and aversion: A synaptic plasticity perspective. *Curr Opin Neurobiol* 35:1145–1157.
26. Vidal-Gadea AG, Pierce-Shimomura JT (2012) Conserved role of dopamine in the modulation of behavior. *Commun Integr Biol* 5:440–447.

27. Menegas W, Bergan JF, Ogawa SK, Isogai Y, Venkataraju KU, Osten P, Uchida N, Watabe-Uchida M (2015) Dopamine neurons projecting to the posterior striatum form an anatomically distinct subclass. *eLife* 4:e10032.
28. Trutti AC, Mulder MJ, Hommel B, Forstmann BU (2019) Functional neuroanatomical review of the ventral tegmental area. *NeuroImage* 191:258–268.
29. Haber SN (2011) Neuroanatomy of reward: A view from the ventral striatum. In: *Neurobiology of Sensation and Reward* (Gottfried JA, ed). Boca Raton, FL: CRC Press/Taylor & Francis.
30. Cai J, Tong Q (2022) Anatomy and function of ventral tegmental area glutamate neurons. *Front Neural Circuits* 16:867053.
31. Zhuang Y, Xu P, Mao C, Wang L, Krumm B, Zhou XE, Huang S, Liu H, Cheng X, Huang X-P, Shen D-D, Xu T, Liu Y-F, Wang Y, Guo J, Jiang Y, Jiang H, Melcher K, Roth BL, Zhang Y, Zhang C, Xu HE (2021) Structural insights into the human D1 and D2 dopamine receptor signaling complexes. *Cell* 184:931–942.e18.
32. Teng X, Chen S, Wang Q, Chen Z, Wang X, Huang N, Zheng S (2022) Structural insights into G protein activation by D1 dopamine receptor. *Sci Adv* 8:eabo4158.
33. Gerfen CR, Engber TM, Mahan LC, Susel Z, Chase TN, Monsma FJ Jr, Sibley DR (1990) D1 and D2 dopamine receptor-regulated gene expression of striatonigral and striatopallidal neurons. *Science* 250:1429–1432.
34. Skeberdis VA, Chevaleyre V, Lau CG, Hemar A, Pertwee RG, Bliss TVP, Malenka RC (2006) Protein kinase A regulates calcium permeability of NMDA receptors. *Nat Neurosci* 9:501–510.
35. Iino Y, Sawada T, Yamaguchi K, et al. (2020) Dopamine D2 receptors in discrimination learning and spine enlargement. *Nature* 579:555–560.
36. Lau GC, Saha S, Faris R, Russek SJ (2004) Up-regulation of NMDAR1 subunit gene expression in cortical neurons via a PKA-dependent pathway. *J Neurochem* 88:564–575.
37. Nayak A, Zastrow D, Lickteig R, Zahniser NR, Browning MD (1998) Maintenance of late-phase LTP is accompanied by PKA-dependent increase in AMPA receptor synthesis. *Nature* 394:680–683.
38. Vaughn AR, Davis MJ, Sivamani RK, Isseroff RR (2018) A concise review of the conflicting roles of dopamine-1 versus dopamine-2 receptors in wound healing. *Molecules* 23:50.
39. Schultz W (2016) Dopamine reward prediction-error signalling: A two-component response. *Nat Rev Neurosci* 17:183–195.
40. Rescorla RA, Wagner AR (1972) Classical conditioning: Current research and theory. New York: Appleton-Century-Crofts.
41. Dayan P, Daw ND (2008) Decision theory, reinforcement learning, and the brain. *Cogn Affect Behav Neurosci* 8:429–453.
42. Katahira K (2015) The relation between reinforcement learning parameters and the influence of reinforcement history on choice behavior. *J Math Psychol* 66:59–69.
43. Schultz W, Dayan P, Montague PR (1997) A neural substrate of prediction and reward. *Science* 275:1593–1599.
44. Eshel N, Tian J, Bukwich M, Uchida N (2016) Dopamine neurons share common response function for reward prediction error. *Nat Neurosci* 19:479–486.

45. Steinberg EE, Keiflin R, Boivin JR, Witten IB, Deisseroth K, Janak PH (2013) A causal link between prediction errors, dopamine neurons, and learning. *Nat Neurosci* 16:966–973.
46. Waelti P, Dickinson A, Schultz W (2001) Dopamine responses comply with basic assumptions of formal learning theory. *Nature* 412:43–48.
47. Yamaguchi K, Maeda Y, Sawada T, Iino Y, Tajiri M, Nakazato R, Ishii S, Kasai H, Yagishita S (2022) A behavioural correlate of the synaptic eligibility trace in the nucleus accumbens. *Sci Rep* 12:1921.
48. Yagishita S, Hayashi-Takagi A, Ellis-Davies GCR, Urakubo H, Ishii S, Kasai H (2014) A critical time window for dopamine actions on the structural plasticity of dendritic spines. *Science* 345:1616–1620.
49. Shouval HZ, Kirkwood A (2025) Eligibility traces as a synaptic substrate for learning. *Curr Opin Neurobiol* 91:102978.
50. Floresco SB, West AR, Ash B, Moore H, Grace AA (2003) Afferent modulation of dopamine neuron firing differentially regulates tonic and phasic dopamine transmission. *Nat Neurosci* 6:968–973.
51. Grace, A. A. (1991). Phasic versus tonic dopamine release and the modulation of dopamine system responsivity: A hypothesis for the etiology of schizophrenia. *Neuroscience* 41, 1–24.
52. Wang Y, Toyoshima O, Kunitatsu J, Yamada H, Matsumoto M (2021) Tonic firing mode of midbrain dopamine neurons continuously tracks reward values changing moment-by-moment. *eLife* 10:e63166.
53. Mikhael JG, Bogacz R (2016) Learning reward uncertainty in the basal ganglia. *PLoS Comput Biol* 12:e1005062.
54. Romero Pinto S, Uchida N (2025) Tonic dopamine and biases in value learning linked through a biologically inspired reinforcement learning model. *Nat Commun* 16:7529.
55. Xu Y, Lin Y, Yu M, Zhou K (2024) The nucleus accumbens in reward and aversion processing: Insights and implications. *Front Behav Neurosci* 18:1420028.
56. Klawonn AM, Malenka RC (2019) Nucleus accumbens modulation in reward and aversion. *Cold Spring Harb Symp Quant Biol* 83:119–129.
57. Yin HH, Ostlund SB, Balleine BW (2008) Reward-guided learning beyond dopamine in the nucleus accumbens: The integrative functions of cortico-basal ganglia networks. *Eur J Neurosci* 28:1437–1448.
58. Hikida T, Yao S, Macpherson T, et al. (2020) Nucleus accumbens pathways control cell-specific gene expression in the medial prefrontal cortex. *Sci Rep* 10:1838.
59. Yawata S, Yamaguchi T, Danjo T, Hikida T, Nakanishi S (2012) Pathway-specific control of reward learning and its flexibility via selective dopamine receptors in the nucleus accumbens. *Proc Natl Acad Sci USA* 109:12764–12769.
60. Lai C-W, Shih C-W, Chang C-H (2022) Analysis of collateral projections from the lateral orbitofrontal cortex to nucleus accumbens and basolateral amygdala in rats. *J Neurophysiol* 127:1535–1546.
61. Xu Y, Lin Y, Yu M, Zhou K (2024) The nucleus accumbens in reward and aversion processing: Insights and implications. *Front Behav Neurosci* 18:1420028.
62. Ibrahim KM, Massaly N, Yoon HJ, et al. (2024) Dorsal hippocampus to nucleus accumbens projections drive reinforcement via activation of accumbal dynorphin neurons. *Nat Commun* 15:750.

63. Castro DC, Bruchas MR (2019) A motivational and neuropeptidergic hub: Anatomical and functional diversity within the nucleus accumbens shell. *Neuropsychopharmacology* 102:529–552.
64. Kravitz AV, Kreitzer AC (2012) Striatal mechanisms underlying movement, reinforcement, and punishment. *Physiology* 27:167–177.
65. Shen W, Flajolet M, Greengard P, Surmeier DJ (2008) Dichotomous dopaminergic control of striatal synaptic plasticity. *Science* 321:848–851.
66. Gerfen CR, Engber TM, Mahan LC, Susel Z, Chase TN, Monsma FJ Jr, Sibley DR (1990) D1 and D2 dopamine receptor-regulated gene expression of striatonigral and striatopallidal neurons. *Science* 250:1429–1432.
67. Macpherson T, Morita M, Hikida T (2014) Striatal direct and indirect pathways control decision-making behavior. *Front Psychol* 5:1301.
68. Lee SJ, Lodder B, Chen Y, et al. (2021) Cell-type-specific asynchronous modulation of PKA by dopamine in learning. *Nature* 590:451–456.
69. Steinberg EE, Boivin JR, Saunders BT, Witten IB, Deisseroth K, Janak PH (2014) Positive reinforcement mediated by midbrain dopamine neurons requires D1 and D2 receptor activation in the nucleus accumbens. *PLoS One* 9:e94771.
70. Morrison JH, Hof PR (1997) Life and death of neurons in the aging brain. *Science* 278:412–419.
71. Goldman-Rakic PS, Brown RM (1981) Regional changes of monoamines in cerebral cortex and subcortical structures of aging rhesus monkeys. *Neuroscience* 6:177–187.
72. Kish SJ, Shannak K, Rajput A, Deck JHN, Hornykiewicz O (1992) Aging produces a specific pattern of striatal dopamine loss: Implications for the etiology of idiopathic Parkinson's disease. *J Neurochem* 58:642–648.
73. Karrer TM, Josef AK, Mata R, Morris ED, Samanez-Larkin GR (2017) Reduced dopamine receptors and transporters but not synthesis capacity in normal aging adults: A meta-analysis. *Neurobiol Aging* 57:36–46.
74. Dang LC, Castellon JJ, Perkins SF, Le NT, Cowan RL, Zald DH, Samanez-Larkin GR (2017) Reduced effects of age on dopamine D2 receptor levels in physically active adults. *NeuroImage* 148:123–129.
75. Fearnley JM, Lees AJ (1991) Ageing and Parkinson's disease: Substantia nigra regional selectivity. *Brain* 114:2283–2301.
76. Rodriguez M, Morales I, Rodriguez-Sabate C, Sanchez A, Castro R, Brito JM, Sabate M (2014) The degeneration and replacement of dopamine cells in Parkinson's disease: The role of aging. *Front Neuroanat* 8:80.
77. Berry AS, Jagust WJ, Hsu M (2019) Age-related variability in decision-making: Insights from neurochemistry. *Cogn Affect Behav Neurosci* 19:415–434.
78. Eppinger B, Schuck NW, Nystrom LE, Cohen JD (2013) Reduced striatal responses to reward prediction errors in older compared with younger adults. *J Neurosci* 33:9905–9912.
79. Dreher J-C, Meyer-Lindenberg A, Kohn P, Berman KF (2008) Age-related changes in midbrain dopaminergic regulation of the human reward system. *Proc Natl Acad Sci USA* 105:15106–15111.
80. Tryon VL, Baker PM, I JM, Rapp PR, Mizumori SJM (2020) Loss of sensitivity to rewards by dopamine neurons may underlie age-related increased probability discounting. *Front Aging Neurosci* 12:49.

81. Lei HC, Parker KE, Kuo C-C, Yuede CM, McCall JG, Imai S-I (2025) Aging reduces motivation through decreased Bdnf expression in the ventral tegmental area. *Mol Psychiatry*.
82. Cools R, Frank MJ, Gibbs SE, Miyakawa A, Jagust W, D'Esposito M (2009) Striatal dopamine predicts outcome-specific reversal learning and its sensitivity to dopaminergic drug administration. *J Neurosci* 29:1538–1543.
83. Rutledge RB, Lazzaro SC, Lau B, Myers CE, Gluck MA, Glimcher PW (2009) Dopaminergic drugs modulate learning rates and perseveration in Parkinson's patients in a dynamic foraging task. *J Neurosci* 29:15104–15114.
84. Chowdhury R, Guitart-Masip M, Lambert C, Dayan P, Huys Q, Düzel E, Dolan RJ (2013) Dopamine restores reward prediction errors in old age. *Nat Neurosci* 16:648–653.
85. Moustafa AA (2011) Levodopa enhances reward learning but impairs reversal learning in Parkinson's disease patients. *Front Hum Neurosci* 4:240.
86. Lodder B, Kamath T, Savenco E, et al. (2025) Absolute measurement of fast and slow neuronal signals with fluorescence lifetime photometry at high temporal resolution. *Neuron* 113:3554–3566.e7.
87. Chen Y, Saulnier JL, Yellen G, Sabatini BL (2014) A PKA activity sensor for quantitative analysis of endogenous GPCR signaling via 2-photon FRET-FLIM imaging. *Front Pharmacol* 5:56.
88. Mastro K, Lee W-C, Wang W, Meng S, Stevens B, Sabatini B (2024) Developmental maturation of frontal cortical circuits impacts decision-making. *bioRxiv*. doi:10.1101/2024.05.24.595609.
89. Beron CC, Neufeld SQ, Linderman SW, Sabatini BL (2022) Mice exhibit stochastic and efficient action switching during probabilistic decision making. *Proc Natl Acad Sci USA* 119:e2113961119.
90. Linker KE, Duran-Laforet V, Ollivier M, Yu X, Schafer DP, Khakh BS (2025) Aging in mice alters regionally enriched striatal astrocytes. *Nat Commun* 16:8496.
91. Liu C, Goel P, Kaeser PS (2021) Spatial and temporal scales of dopamine transmission. *Nat Rev Neurosci* 22:345–358.
92. Long KLP, Muroy SE, Sorooshyari SK, Ko MJ, Jaques Y, Mohan K, Sudmant P, Kaufer D (2025) Transcriptomic profiles of susceptibility and resilience to stress in the amygdala and hippocampus of male rats. *Neurobiol Stress* 38:100754.
93. Kilfeather P, Khoo JH, Wagner K, Liang H, Caiazza MC, An Y, Zhang X, Chen X, Connor-Robson N, Shang Z, Wade-Martins R (2024) Single-cell spatial transcriptomic and translational profiling of dopaminergic neurons in health, aging, and disease. *Cell Rep* 43:113784.

VI. Supplementary

A. Statistical Supplementary

All statistical tests, comparisons, and their presentation status in the main text

Key

Symbol / Shading	Meaning
✓ (green highlight)	Comparison selected for presentation in main text or figure
n.s.	Not significant — reported explicitly as null result
No highlight	Significant but not the primary comparison highlighted in the paper
Not presented	Test run for completeness; not reported in main text or figure

Figure 1. Age-dependent differences in acquisition of deterministic task structure

n = 8, 27, 9, 17, 8, 11 mice for age groups 0–2, 2–4, 4–6, 6–12, 12–18, 18–24 months. All pairwise tests: Tukey HSD post-hoc following one-way ANOVA unless stated.

Fig 1b. Acquisition speed (sessions to criterion)

One-way ANOVA: $F(5,74) = 17.06$, $p < 0.0001$

Comparison	Mean Diff (sessions)	p-adj (Tukey)	Presented in Paper
0–2 vs 12–18	7.88	< 0.001	
0–2 vs 18–24	9.03	< 0.001	
0–2 vs 4–6	4.26	0.010	✓ (selected)
0–2 vs 6–12	6.89	< 0.001	
18–24 vs 2–4	-6.03	< 0.001	
18–24 vs 4–6	-4.76	0.001	

12–18 vs 2–4	-4.88	< 0.001	
12–18 vs 4–6	-3.61	0.045	✓ (selected)
2–4 vs 6–12	3.89	< 0.001	

Fig 1c — Rewarded and unrewarded trials across learning phases

Early phase. Unrewarded trials [ANOVA: $F(5,74) = 6.37$, $p < 0.001$]

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0–2 vs 18–24	114.5	0.001	
12–18 vs 18–24	95.3	0.007	
18–24 vs 2–4	-103.7	< 0.001	✓
18–24 vs 4–6	-114.0	< 0.001	
18–24 vs 6–12	-96.7	0.001	

Early phase. Rewarded trials [ANOVA: $F(5,74) = 7.80$, $p < 0.001$]

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0–2 vs 18–24	214.05	0.0004	
18–24 vs 2–4	-199.2167	< 0.001	✓
18–24 vs 4–6	-229.1636	< 0.001	

18–24 vs 6–12	-148.0667	0.0078	
---------------	-----------	--------	--

Middle phase. Unrewarded trials [ANOVA: $F(5,74) = 3.62$, $p = 0.006$]

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0–2 vs 18–24	121.9	0.042	✓
18–24 vs 4–6	-148.8	0.002	✓

Middle phase. Rewarded trials [ANOVA: $F(5,74) = 1.14$, $p = 0.345$]

Not significant. Not presented.

Late phase. Rewarded and Unrewarded trials

Late rewarded: ANOVA $F(5,70) = 0.84$, $p = 0.526$ — not significant. Late unrewarded: ANOVA $F(5,70) = 3.29$, $p = 0.010$ — significant overall but no individual Tukey comparisons reached significance. Consistent with convergence of age groups in later learning. Not highlighted.

Fig 1d. Reward rate across learning phases

Early phase [ANOVA: $F(5,132) = 7.48$, $p < 0.001$]

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0–2 vs 18–24	-0.262	< 0.001	
12–18 vs 18–24	-0.194	0.005	
18–24 vs 2–4	0.213	< 0.001	✓
18–24 vs 4–6	0.262	< 0.001	
18–24 vs 6–12	0.194	0.001	

Middle phase [ANOVA: $F(5,284) = 1.82$, $p = 0.110$]

Not significant. Not presented.

Late phase [ANOVA: $F(5,221) = 1.16$, $p = 0.330$]

Not significant. Not presented.

Fig 1e. Switch probability across learning phases

Early: $F(5,132) = 0.97, p = 0.439$ | Middle: $F(5,284) = 0.75, p = 0.585$ | Late: $F(5,221) = 0.86, p = 0.510$. No significant differences at any phase. Reported in text as null result confirming preserved exploratory switching.

Fig 1f. Trial duration across learning phases

Early phase [ANOVA: $F(5,132) = 4.74, p < 0.001$]

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
12–18 vs 2–4	-0.211	0.022	✓
2–4 vs 4–6	0.232	0.004	
4–6 vs 6–12	-0.223	0.013	

Middle phase [ANOVA: $F(5,284) = 9.46, p < 0.001$]

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0–2 vs 12–18	0.109	0.007	
0–2 vs 18–24	0.111	0.004	
0–2 vs 4–6	0.088	0.036	
12–18 vs 2–4	-0.106	< 0.001	✓
18–24 vs 2–4	-0.108	< 0.001	✓
18–24 vs 6–12	-0.065	0.048	
2–4 vs 4–6	0.085	< 0.001	

Late phase [ANOVA: $F(5,221) = 24.95$, $p < 0.001$]

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0-2 vs 12-18	0.110	< 0.001	✓
0-2 vs 18-24	0.087	< 0.001	✓
0-2 vs 4-6	0.075	< 0.001	
0-2 vs 6-12	0.072	< 0.001	
12-18 vs 2-4	-0.105	< 0.001	
18-24 vs 2-4	-0.083	< 0.001	
2-4 vs 4-6	0.070	< 0.001	✓
2-4 vs 6-12	0.067	< 0.001	

Trial duration differences in figure highlight that duration increases early but stabilises across older age groups. Key interpretation: duration does not track error rates, indicating preserved task engagement.

Figure 2. Aging impairs reinforcement-dependent updating during probabilistic decision-making

$n = 8, 24, 11, 15, 8, 9$ mice for age groups 0-2, 2-4, 4-6, 6-12, 12-18, 18-24 months.

Fig 2a. Rewarded trials

One-way ANOVA: $F(5,69) = 3.76$, $p = 0.005$ | Kruskal-Wallis: $H = 15.07$, $p = 0.010$

Significant differences driven by 0-2 month group having more rewarded trials than all others.

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0-2 vs 2-4	-59.5944	0.0047	✓

Fig 2a. Unrewarded trials

One-way ANOVA: $F(5,69) = 12.01$, $p < 0.001$ | Kruskal-Wallis: $H = 29.60$, $p < 0.001$

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
------------	--------------------	---------------	-----------

0-2 vs 12-18	-51.6	0.002	
0-2 vs 2-4	-59.0	< 0.001	✓
0-2 vs 4-6	-74.2	< 0.001	
0-2 vs 6-12	-62.5	< 0.001	
18-24 vs 2-4	-37.5	0.004	
18-24 vs 4-6	-52.6	< 0.001	✓
18-24 vs 6-12	-41.0	0.003	✓

Fig 2b. Trial duration

One-way ANOVA: $F(5,69) = 3.99$, $p = 0.003$ | Kruskal-Wallis: $H = 32.19$, $p < 0.001$

Mixed linear model confirms age-related differences (all groups vs 0-2 months significant). Key finding: duration stabilises from 4 months onward — not increasing with error rates.

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
0-2 vs 12-18	0.051	< 0.001	
0-2 vs 18-24	0.049	< 0.001	
0-2 vs 4-6	0.054	< 0.001	
0-2 vs 6-12	0.052	< 0.001	

12–18 vs 2–4	-0.029	0.026	
18–24 vs 2–4	-0.028	0.028	
2–4 vs 4–6	0.032	0.003	✓
2–4 vs 6–12	0.031	0.001	

Fig 2c. High-port selection probability

One-way ANOVA: $F(5,69) = 3.99$, $p = 0.003$

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
18–24 vs 2–4	-0.057	0.009	✓
18–24 vs 4–6	-0.076	0.001	
18–24 vs 6–12	-0.047	0.082	n.s.

Fig 2d, 2e, 2f, 2g. Non-significant measures

The following measures did not yield significant results. Directional trends are discussed in the main text as consistent with the overall aging narrative, with the caveat that these may reflect limited statistical power given current sample sizes.

Measure	ANOVA statistic	p-value	Status
Pre-switch pHigh	$F(5,69) = 2.14$	0.071	Not presented
Post-reversal recovery	$F(5,68) = 0.70$	0.623	Not presented
Post-reversal switch prob.	$F(5,69) = 1.20$	0.319	Not presented
Lose-switch probability	$F(5,69) = 1.58$	0.177	Not presented
Win-stay probability	$F(5,69) = 0.86$	0.512	Not presented

Figure 3. Model reveals age-dependent reduction in reinforcement sensitivity and prolonged persistence of prior evidence

n = 8, 24, 11, 15, 8, 10 mice for age groups 0–2, 2–4, 4–6, 6–12, 12–18, 18–24 months.

Summary of RFLR model parameter statistics

Parameter	ANOVA statistic	p-value	Presentation status
Alpha (persistence)	$F(5,70) = 1.43$	0.224	Not significant, not highlighted
Beta (reinf. sensitivity)	$F(5,70) = 2.68$	0.029	✓ Significant, highlighted
Tau (integration timescale)	$F(5,70) = 3.10$	0.014	✓ Significant, highlighted
Model fit quality (LLS)	$F(5,70) = 0.87$	0.504	Not significant, reported as validation

Fig 3d. Beta (reinforcement sensitivity) post-hoc comparisons

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
18–24 vs 4–6	0.704	0.023	✓

Fig 3e. Tau (integration timescale) post-hoc comparisons

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
18–24 vs 2–4	-2.110	0.007	✓
18–24 vs 4–6	-2.238	0.017	✓

Fig 3f. Model fit quality

One-way ANOVA: $F(5,70) = 0.87$, $p = 0.504$

Spearman correlation with age: $\rho = -0.037$, $p = 0.753$

Reported as validation that the model provides a comparably good description of behavior across all age groups. No significant differences, this is the desired result.

Figure 4. Dopamine transients encode reward history in deterministic and probabilistic versions of the task

No formal inferential statistics presented for Figure 4. Values reported as mean $\pm$ SEM. Differences between age groups and across sequence positions are described qualitatively in the main text.

Peak z-score values (mean $\pm$ SEM)

100/0 contingency, Aligned to consumption

Group	Sequence 1	Sequence 2	Sequence 3
P120	3.82 ± 0.05	3.57 ± 0.05	3.34 ± 0.06
P60	3.36 ± 0.06	2.84 ± 0.07	2.74 ± 0.07

100/0 contingency, Aligned to cue

Group	Sequence 1	Sequence 2	Sequence 3
-------	------------	------------	------------

P120	1.82 ± 0.05	2.39 ± 0.05	2.34 ± 0.05
P60	1.91 ± 0.05	2.28 ± 0.05	2.22 ± 0.05

100/0 contingency, Loss sequence aligned to consumption

Group	Sequence 1	Sequence 2	Sequence 3
P120	-1.27 ± 0.02	-1.31 ± 0.03	-1.26 ± 0.03
P60	-1.47 ± 0.03	-1.41 ± 0.03	-1.36 ± 0.04

80/20 contingency, Aligned to consumption

Group	Sequence 1	Sequence 2	Sequence 3
P120	4.05 ± 0.05	3.28 ± 0.04	3.11 ± 0.05
P60	3.60 ± 0.05	2.50 ± 0.05	2.36 ± 0.06

Figure 5. Aging produces a downward shift in tonic dopamine baseline while preserving relative phasic signalling

n = 10, 6, 14 recordings from 4, 3, and 5 mice at 3–6, 18–21, and 24+ months respectively. Several animals contributed recordings at multiple timepoints (longitudinal design).

Fig 5d. Baseline lifetime (modal signal)

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
18–21 mo vs 24+ mo	-0.201	0.001	✓
24+ mo vs 3–6 mo	0.1288	0.013	✓
18–21 mo vs 3–6 mo	-0.0718	0.269	n.s.

Fig 5e. Peak lifetime

One-way ANOVA: $F(2,21) = 9.47$, $p = 0.001$

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
18–21 mo vs 24+ mo	-0.161	0.001	✓
24+ mo vs 3–6 mo	0.107	0.013	✓
18–21 mo vs 3–6 mo	-0.054	0.3287	n.s.

Fig 5f. Phasic amplitude (peak minus baseline)

No significant differences between age groups (all $p > 0.05$).

One-way ANOVA: $F(2,21) = 2.97$, $p = 0.073$

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
18–21 mo vs 24+ mo	0.0393	0.063	n.s.
24+ mo vs 3–6 mo	-0.0217	0.3054	n.s.
18–21 mo vs 3–6 mo	0.0176	0.5098	n.s.

Fig 5g. Autofluorescence control

Fluorescence intensity scales linearly with excitation power across all animals ($R^2 = 0.977$ – 0.998 for representative 24 month animals). Confirms sensor-dependent emission.

Figure 6. Tonic dopamine baseline does not predict motivational effort or hunger state

Fig 6b–c: $n = 34$ recordings from 13 mice. Fig 6d: $n = 8$ animals. Fig 6e: see per-group session counts below.

Fig 6b. Breakpoint vs body weight

Mixed effects model: $\beta = -6.56$, $SE = 4.41$, $p = 0.137$ (not significant)

One-way ANOVA by weight bin: $F(3,30) = 4.65$, $p = 0.009$ | Kruskal-Wallis: $H = 12.90$, $p = 0.005$

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
80–85% vs 90–95%	-151.2	0.009	✓
80–85% vs 85–90%	-67.3	0.249	n.s.

Fig 6c. Breakpoint across age groups

One-way ANOVA: $F(2,30) = 1.38$, $p = 0.268$ | Kruskal-Wallis: $H = 4.72$, $p = 0.194$

Comparison	Mean Diff (trials)	p-adj (Tukey)	Presented
3 mo vs 12 mo	-81.5	0.234	n.s.
3 mo vs 18 mo	-24.5	0.969	n.s.
3 mo vs 24 mo	-59.0	0.627	n.s.

Fig 6d. Dopamine lifetime vs PR breakpoint

Measure	Correlation	p-value	Conclusion
Baseline DA vs Breakpoint	$r = 0.005$	0.991	Not significant
Peak DA vs Breakpoint	$r = -0.183$	0.665	Not significant

Fig 6e. Baseline dopamine lifetime: sated vs hungry

3–6 months: $n = 10$ sated, 12 hungry and $p = 0.176$ (Mann–Whitney U test)

18–21 months: $n = 10$ sated, 7 hungry and $p = 0.364$ (Mann–Whitney U test)

24+ months: n = 12 sated, 3 hungry and p = 0.136 (Mann–Whitney U test)

No significant difference between hunger states in any age group. Reported as null result confirming that tonic dopamine baseline is not acutely modulated by hunger.

Figure 7. Aging drives transcriptional changes in striatal interneurons without altering canonical marker expression

F(

Fig 7e. Astrocyte contamination marker expression across age
Mann-Whitney U test, two-sided. Young n = 436, Aged n = 105.

Gene	p-value	Sig
Slc1a2	2.26e-12	***
Vim	1.93e-02	*
S100b	2.43e-02	*
Aldoc	—	n.s.

Fig 7g. Interneuron subtype marker expression across age (Striatal Interneuron cluster)
Mann-Whitney U test, two-sided. Young n = 438, Aged n = 726.

Gene	p-value	Significance
Sst	3.72e-05	***
Npy	4.59e-06	***
Kcnq3	4.58e-02	*
Cnr1	3.27e-02	*
Chat	1.0000	n.s.
Pvalb	6.02e-01	n.s.
Hcn1, Gabra1, Th	—	n.s.

Sst and Npy show significant age-related reductions in the striatal interneuron population. Canonical cholinergic (Chat), fast-spiking (Pvalb), and other interneuron markers (Hcn1, Gabra1, Th) are not significantly altered by age. Astrocyte markers Slc1a2, Vim, and S100b show elevated expression in the aged subcohort; Aldoc does not differ significantly.